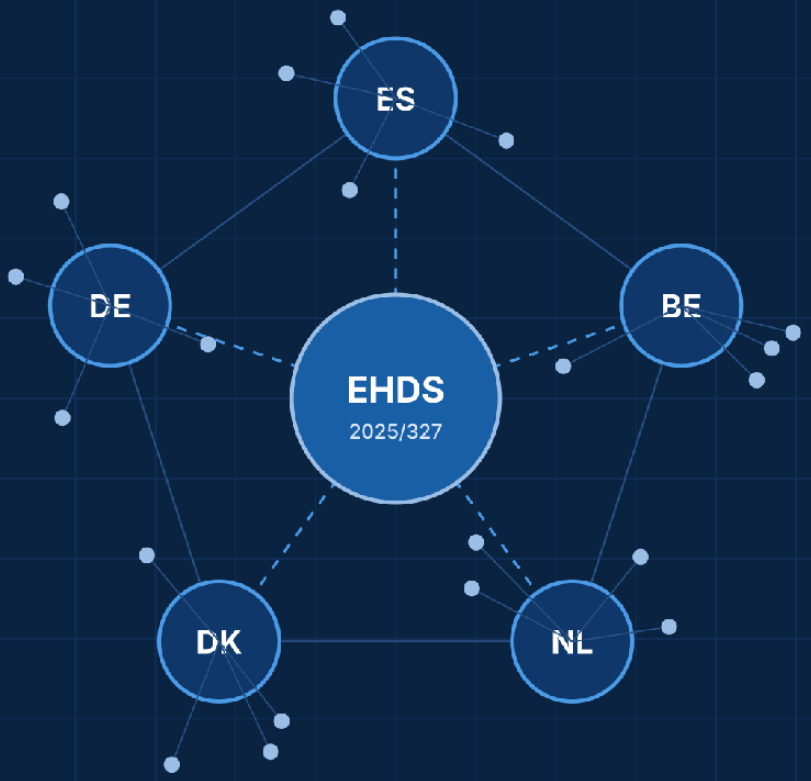
SOFTWARE IN A BOX

Federated data spaces · Madrid · Brussels



COMPARATIVE STUDY · NOVEMBER 2026

Dental health data spaces in Europe

Spain, Belgium, the Netherlands, Denmark and Germany: markets, regulation and alignment with the European Health Data Space (EHDS)

Market needs · Regulatory needs · Use cases · Partner roadmap

Software in a Box

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Document for partners and customers

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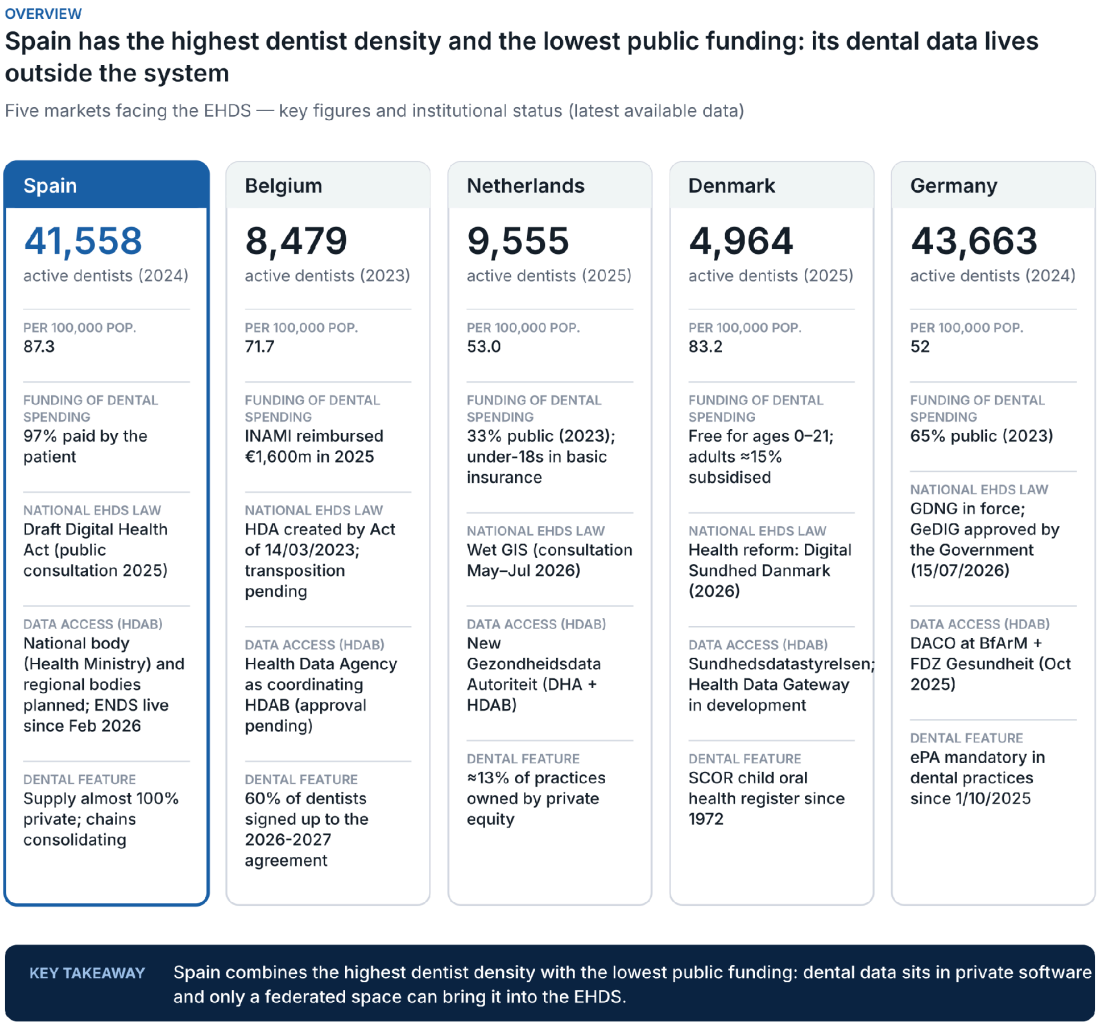
Executive summary

This study compares how five European markets —Spain, Belgium, the Netherlands, Denmark and Germany— are preparing for oral health data to become part of the European Health Data Space (EHDS), created by Regulation (EU) 2025/327. The aim is practical: to identify in each country the market needs and regulatory needs that a federated dental health data space can address, and to align them with the obligations and opportunities the EHDS introduces between 2027 and 2031.

The central conclusion is that **dental data is today the great absentee from Europe's health data architecture**, not for lack of clinical or economic value, but because of the way dentistry is organised in Europe: small practices, mostly private spending for adults, proprietary practice software and little integration with public health records. The EHDS does not solve this on its own; indeed, by exempting microenterprises from data holder obligations unless a Member State decides otherwise [1], it may leave most clinics out. A federated data space that aggregates clinics, chains and insurers is the most direct way for oral health data to reach health data access bodies (HDABs) without professionals giving up control of their records.

Five very different starting points

- **Spain** combines the highest dentist density in the group (87.3 per 100,000 inhabitants) with spending that patients pay 97% of themselves [2]. It already has a digital national health record and interoperable ePrescription in place, and a National Health Data Space live since February 2026 with more than 330 datasets [3], but the private dentist is outside that circuit. Adaptation to the EHDS will be channelled through the draft Digital Health Act, under public consultation since September 2025 [4].
- **Belgium** has a public reimbursement system (INAMI) that turns every reimbursable dental act into structured data: INAMI reimbursed €1.6 billion in dental care in 2025 [5]. Its Health Data Agency, created by law in 2023, already maintains a national catalogue of more than 480 datasets and is set to become the coordinating HDAB [6].
- **The Netherlands** suffers a structural shortage of dentists —a quarter were trained abroad [7]— and rapid consolidation: more than 500 practices owned by private equity, accounting for about 30% of revenue [8]. The Wet GIS creates a single health data authority that will take on the roles of digital health authority and HDAB [9].
- **Denmark** offers free dental care up to age 21 and has kept the SCOR child oral health register since 1972, a European example of public oral health data used to make policy [10][11]. The challenge is adults, who are subsidised only around 15%.
- **Germany** is the most advanced country: the ePA has been mandatory in dental practices since 1 October 2025, there are more than 5.5 million electronic dental bonus booklets (eZahnbonusheft) in the ePA [12] and the Forschungsdatenzentrum Gesundheit has been operating since 9 October 2025 [13].



Infographic 1 · *Five markets facing the EHDS: key figures and institutional status*

What this means for Software in a Box and its partners

The analysis allows commercial entry to be prioritised. **Spain and the Netherlands** are the markets where a private federated space adds the most value, because data is fragmented in private software and there is strong private pressure —chains, insurers, funds— to share and compare data. **Germany** is a complement market: you do not compete with the ePA, but sell data quality, compliance and federated analytics to chains, investor-owned centres (iMVZ) and software vendors. **Belgium and Denmark** are institutional alliance markets, where the value lies in projects with the data authority to bring in adults and private clinics.

In every case the vehicle is the same: a data space connector built into clinical software and deployed on Microsoft Azure by integrators and partners, allowing each clinic to contribute data to specific use cases —quality across clinics, artificial intelligence on dental imaging, evaluation of prevention programmes, evidence for industry— without moving health records out of its control.

> The window of opportunity is concrete: between the designation of the access bodies (26 March 2027) and the start of most secondary use rules (26 March 2029) there are two years to connect clinics and chains. Whoever reaches 2029 with an operational federated dental space will be the natural counterpart of the HDABs.

Main recommendations

- Close two pilots with dental chains before March 2027 —one in Spain and one in the Netherlands— through Azure partners, focused on comparable quality indicators across clinics.
- Design the connector from the start with the European electronic health record exchange format (EEHRxF) and HealthDCAT-AP metadata, so that every dental dataset can be catalogued by HDABs.
- Propose to chains and insurers the trusted data holder status provided for in Article 72 of the Regulation, which reduces the burden of each access request [14].
- Position the dental imaging module (X-rays and CBCT) as a product line for 2029-2031, when medical images join the priority exchange categories.
- In Germany, enter through a dental practice software vendor; in Belgium and Denmark, through projects with the Health Data Agency and Sundhedsdatastyrelsen.

1. Purpose, scope and methodology

1.1. Purpose of the study

The study answers three questions. First: how are the dental clinic markets of Spain, Belgium, the Netherlands, Denmark and Germany similar and different, and what constraints do they place on an oral health data space? Second: what national rules govern health data in each country today and how are they adapting to the EHDS? Third: which market needs and regulatory needs can be linked to specific mechanisms of Regulation (EU) 2025/327, so that a federated dental data space is presented as a compliance tool and not just a technology project?

The document is intended for three audiences: the Software in a Box team; Microsoft Azure partners and technology integrators who can add the solution to their offering; and sector stakeholders —dental chains, insurers, professional colleges and associations— with whom those partners will work.

1.2. Scope

The analysis covers five dimensions per country: structure of the health system and funding of dental care; structure of supply (professionals, practices and chains); digital health infrastructure and the degree of dentist integration; national regulatory framework and status of adaptation to the EHDS, including the data access body; and the approach —explicit or implicit— to oral health data in public policy.

The analysis of treatment prices and fees has been deliberately excluded, as it does not affect data architecture, as has the detail of professional regulation (specialties, continuing education) except where it affects data availability.

1.3. Methodology and sources

The study is based on public sources consulted in November 2026: the text of the Regulation and European Commission documentation; the country chapters of the Council of European Dentists (CED) Manual of Dental Practice 2026, which provide consistent figures on professionals and funding; official national sources (ministries, statistical institutes, health data agencies, public insurers and parliaments); analysis by specialised law firms; and trade and business press for market data (consolidation, chains, transactions). References are cited using the notation [Rn] and listed in Annex D.

Where sources give figures for different years, the reference year is indicated. Comparisons between countries should be read with caution: public funding indicators are not homogeneous (share of spending, total reimbursement or coverage by age group) and are presented as published by the sources.

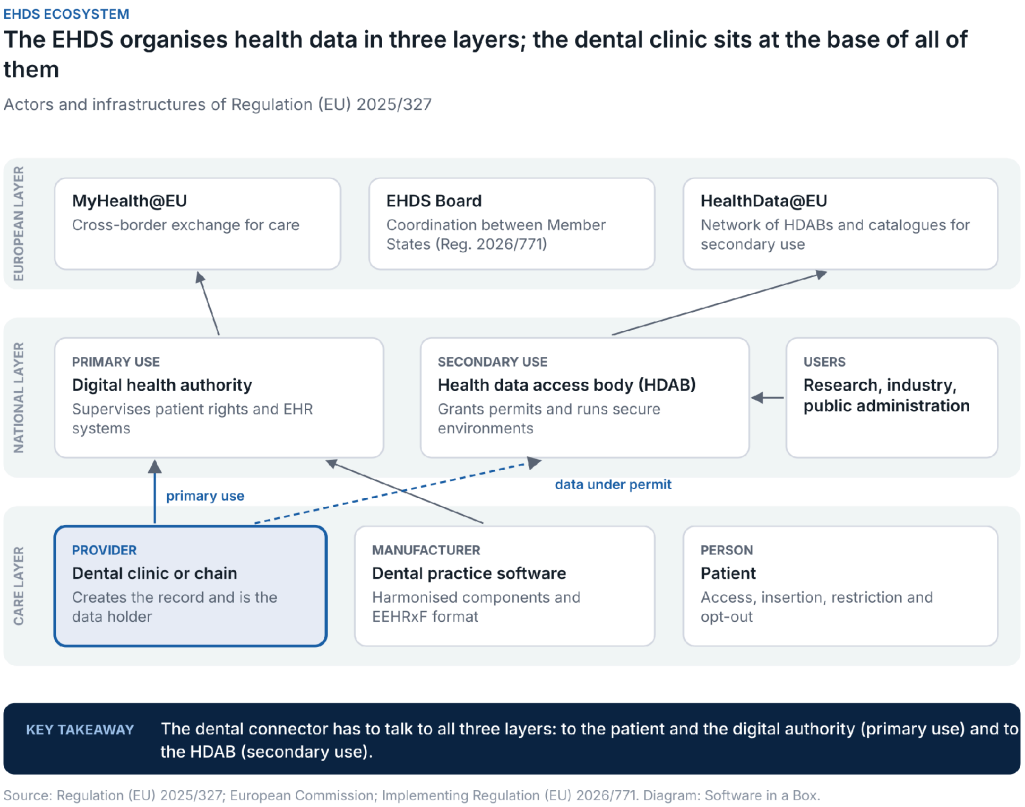
> The readiness assessments by country and the opportunity matrix (chapter 8), the location of dental data by country, the risk matrix and the roadmap are Software in a Box analytical work and not official measurements. They are flagged as such in each piece.

2. The European Health Data Space: what changes for dentistry

2.1. What the EHDS is

Regulation (EU) 2025/327 of 11 February 2025 on the European Health Data Space was published in the Official Journal of the EU on 5 March 2025 and entered into force on 26 March 2025 [15][16]. It is the Union's first common data space dedicated to a specific domain and stems from the European Strategy for Data [17]. Its architecture rests on two pillars:

- **Primary use**: the use of electronic health data to provide healthcare. The Regulation strengthens people's rights —to access their data immediately and free of charge, to insert information, to restrict access by professionals and to know who has accessed it— and requires certain priority categories of data to be exchanged in a common European format [15].
- **Secondary use**: the reuse of that data for research, innovation, policy-making, statistics or training, through health data access bodies (HDABs) that grant permits and require data to be processed in secure environments [15][14].

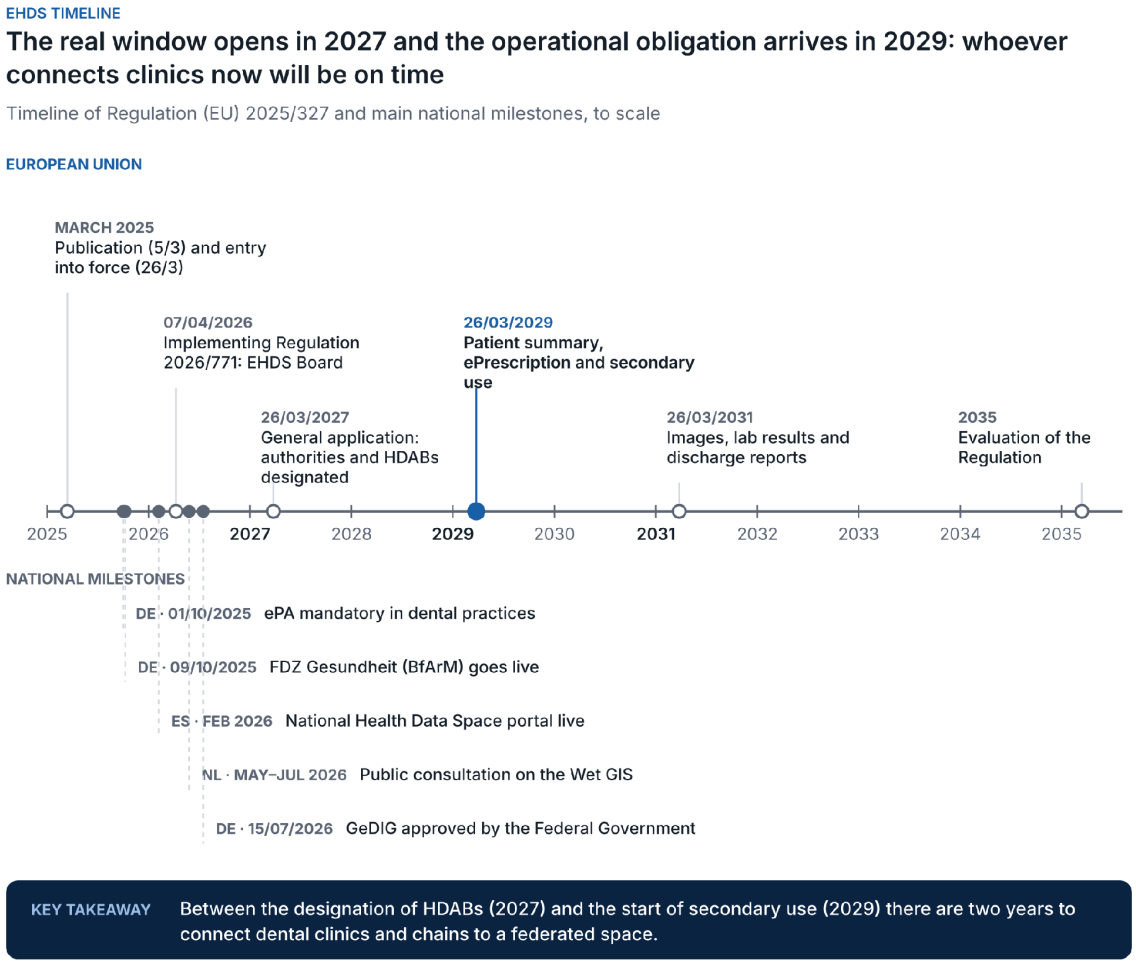


Infographic 2 · *Actors and infrastructures of the EHDS in three layers*

A third block is added, less visible but decisive for the dental sector: a harmonised framework for **electronic health record (EHR) systems**, with common interoperability and logging components and a conformity regime for manufacturers [17].

2.2. A phased timeline

The Regulation applies in phases. The general date of application is **26 March 2027**, by which Member States must have designated their digital health authorities and HDABs [16]. On **26 March 2029** the exchange of the first group of priority categories begins —patient summaries and ePrescription and eDispensation— together with most secondary use rules [17][3]. On **26 March 2031** medical images, test results and discharge reports are added [18]. In parallel, on 7 April 2026 the Commission adopted Implementing Regulation (EU) 2026/771, which governs the functioning of the EHDS Board [19].



Infographic 3 · *EHDS timeline and national milestones to scale*

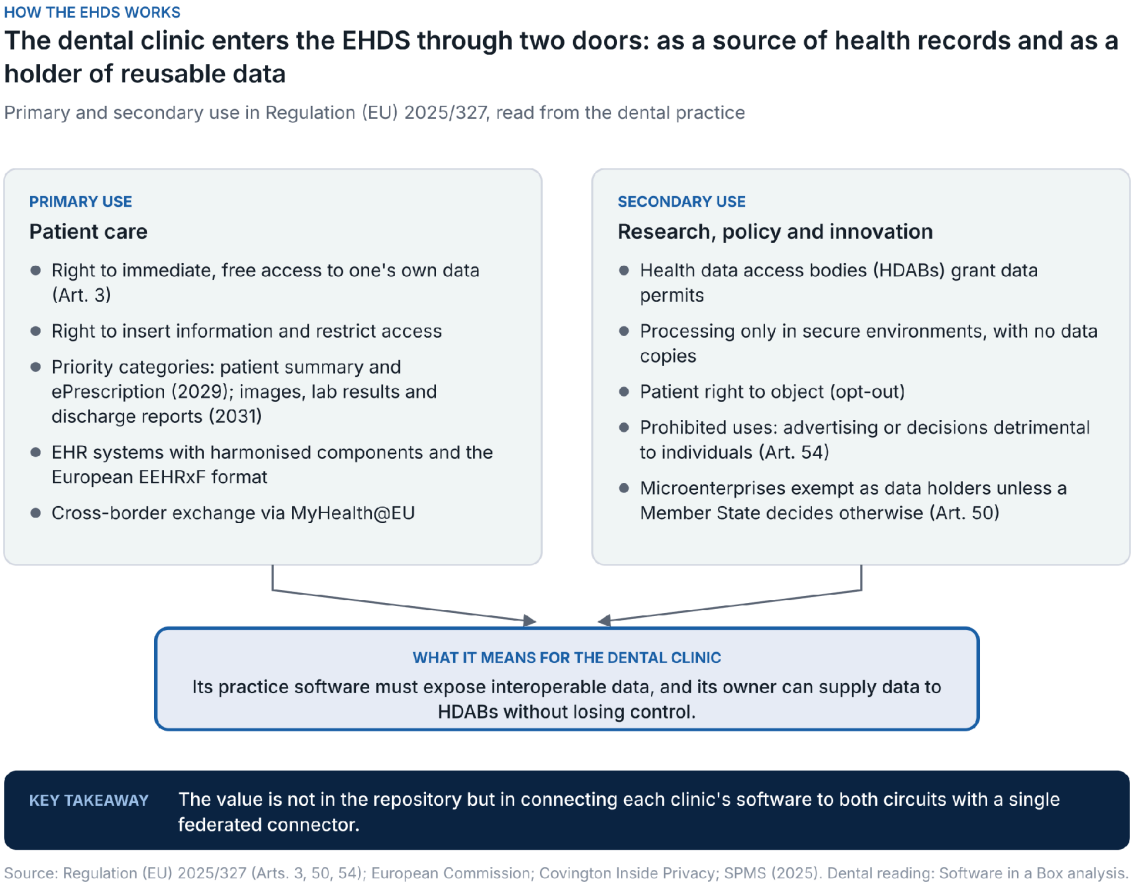
Table 1 · EHDS milestones and their implications for the dental sector

Date	Milestone of Regulation (EU) 2025/327	Implication for clinics, chains and partners
05/03/2025	Publication in the Official Journal of the EU [15]	The transition period begins; the text is the reference for product design.
26/03/2025	Entry into force [16]	Member States start preparing national rules and bodies.

Date	Milestone of Regulation (EU) 2025/327	Implication for clinics, chains and partners
07/04/2026	Implementing Regulation (EU) 2026/771 on the EHDS Board [19]	First piece of European governance; more technical implementing acts will follow.
26/03/2027	General date of application: digital health authorities and HDABs designated [16]	Formal counterparts in each country; the moment to present the dental space as a source.
26/03/2029	Exchange of patient summaries and ePrescriptions; application of most secondary use rules [17][3]	The dental space must be catalogued and ready for access requests.
26/03/2031	Medical images, test results and discharge reports [18]	Dental imaging (X-rays, CBCT) becomes an exchange asset.
2035	Evaluation of the Regulation [16]	Possible review of categories and obligations.

2.3. The dental clinic's two entry doors

The dental clinic relates to the EHDS in two ways. As a **healthcare provider**, it generates health records that patients are entitled to consult and which, insofar as they include priority categories, must be exchangeable in the European format. As a **data holder**, it may be called upon to make data available to HDABs for secondary uses, with the safeguards of the Regulation.



Infographic 4 · *Primary and secondary use read from the dental practice*

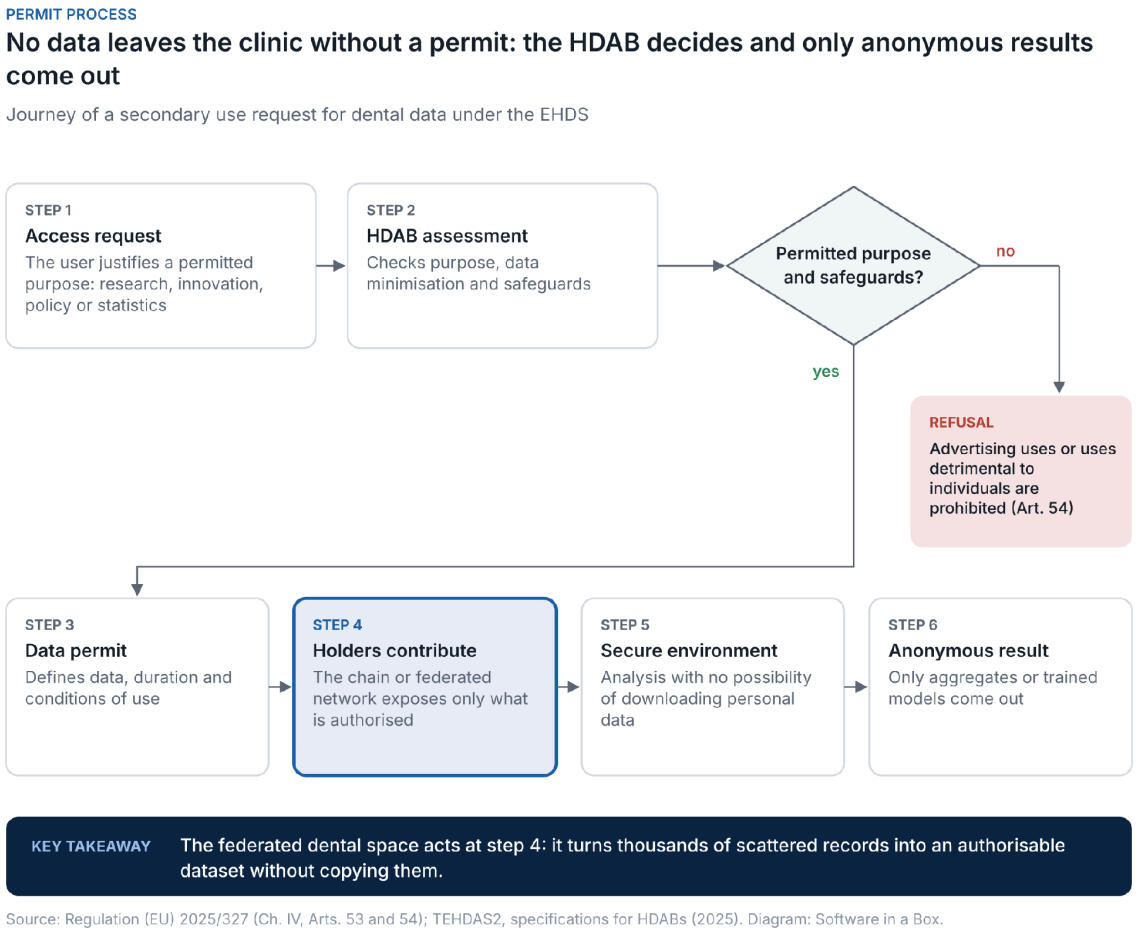
Three elements are particularly relevant for dentistry:

- **Priority categories.** The dental record is not a category in its own right, but several of its contents fit the planned categories: the patient summary (allergies, medication, history relevant to dental treatment),

ePrescription and, from 2031, medical images. In our reading, intraoral X-rays, panoramic radiographs and cone-beam computed tomography (CBCT) are medical images for the purposes of the Regulation, which will turn dental imaging into an asset for exchange and reuse.

- **EHR systems.** Dental practice software that processes priority categories will be subject to the interoperability and logging requirements for EHR systems. In the Netherlands, the forthcoming Wet GIS replaces third-party certification with manufacturer self-assessment in a European testing environment [9], which illustrates where conformity is heading.
- **Prohibited uses.** Article 54 prohibits using data obtained through secondary use for advertising or for decisions detrimental to individuals [14]. For a sector with a strong commercial component, this prohibition is both a limit and an argument for trust.

The diagram below summarises how a piece of dental data reaches a user under the EHDS: no data leaves the clinic without a permit from the access body and, once granted, the analysis takes place in a secure environment from which only anonymous or aggregated results come out.

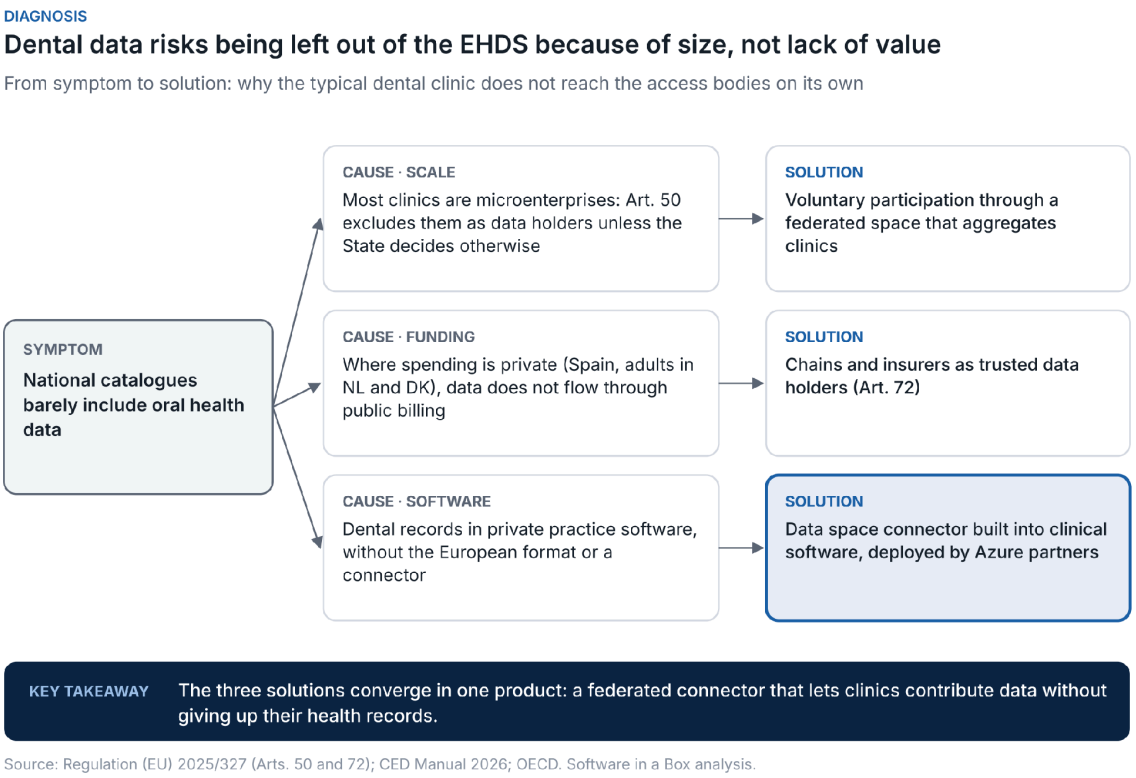


Infographic 5 · *Journey of a secondary use request for dental data*

2.4. The clause that may leave the dentist out: Article 50

Article 50 provides that data holder obligations regarding secondary use do not apply to natural persons or microenterprises, unless the Member State decides to extend them in its national legislation [1][14]. Most European dental practices are small: in the Netherlands, for example, a Rabobank sector analysis from the last decade estimated that 58% of practices were single-handed [8]. If States do not use the option to extend the obligations, most dental data will not reach the HDABs through the mandatory route.

The Regulation does, however, offer two routes that a federated space can use. The first is voluntary participation: nothing prevents exempt clinics from making data available through an intermediary. The second is the **trusted data holder** (Article 72), which allows a designated entity to handle access requests that concern only its own data, easing the burden on the HDAB [14]. For a dental chain with hundreds of clinics, or an insurer with its own network, it is a particularly suitable route.



Infographic 6 · *Why dental data may be left out of the EHDS and how to prevent it*

2.5. Implementation status

One year after entry into force, all Member States are working on the designation of their bodies. The guidelines of the TEHDAS2 Joint Action —on fees, penalties, data access and data access application management systems (DAAMS)— have been put to consultation in successive waves in 2025 and 2026 [9]. The five countries analysed show very different levels of progress, detailed in the following chapters: Germany already operates a research data centre and an access coordination office; Belgium has its agency created by law and an operational catalogue; Denmark has reorganised its digital governance into a new entity; the Netherlands has consulted on the law creating its authority; and Spain is preparing a Digital Health Act while its National Health Data Space is already running.

3. Spain

3.1. Health system and funding of dental care

Spain has a universal National Health System (SNS), funded by taxes and managed by the autonomous communities. Dentistry is the great exception to that universality: public coverage is essentially limited to emergencies, extractions and basic care, and the CED estimates that **97% of dental care is paid directly by the**

patient [2]. Voluntary insurance covering additional dental care reaches only 1-2% of the population, although one in five people has some form of private health insurance [2].

The Oral Health Plan approved in June 2022, with a budget of €68 million, extended the common benefit package for all beneficiaries to some services —treatment of acute infections, extractions— and expanded benefits for children aged 0 to 14, pregnant women and people with disabilities or head and neck cancer [2]. It is a significant step, but it does not change the essentially private nature of the sector.

3.2. Structure of the dental clinic market

Spain has the most dentists in the group analysed, in both absolute and relative terms. According to the INE, there were **43,672 registered dentists in 2025**, 1.9% more than the previous year; 59.9% are women and the highest concentrations are in Madrid (9,707), Catalonia (6,507) and the Valencian Community (4,852) [20]. The CED puts the number of active dentists at 41,558 in 2024, equivalent to **87.3 per 100,000 inhabitants** [2]. The General Council of Dentists warns of possible oversupply, with about 2,000 new graduates a year [20].

Supply is almost entirely private: an analysis by the University of Granada put private practice at around 97%, in more than 23,000 registered clinics [21]. Alongside the independent practice, which remains the majority, the market has undergone intense consolidation. **DONTE Group** —owned by the Canadian fund OTPP— ended 2025 with **440 owned clinics**, 407 of them under the Vitaldent brand, and almost 750,000 diagnoses performed [22]. Health insurers (Sanitas, Adeslas, Asisa, DKV, Caser) have developed their own dental networks, and operators such as Vivanta and Institutos Odontológicos complete a chain segment that is already a major stakeholder.

3.3. Digital health infrastructure

In public healthcare Spain starts from a favourable position: the SNS digital health record and interoperable ePrescription form a base that is already deployed, and all autonomous communities are connected to MyHealth@EU or in testing [3]. The **National Health Data Space (ENDS)** has had its own portal since February 2026, with more than 330 datasets published in its catalogue and a federated model in which sources keep control and data are provided on request in secure environments [3].

The problem for dentistry is that all this infrastructure belongs to the public system. The dental records of the vast majority of patients reside in the practice software of private clinics and chains, with no connection to the SNS health record or the ENDS. The challenge of public-private interoperability is explicitly listed among the five challenges the Government's own data portal identifies ahead of 2029 [3].

3.4. Regulatory framework and adaptation to the EHDS

On 22 September 2025 the Ministry of Health opened the prior public consultation on the **draft Digital Health Act**, which will adapt Spanish law to the EHDS Regulation and regulate the interoperable digital health record and the use of digital technologies in care [4]. The consultation proposed regulating an interoperable health record across the whole country and **including private providers** [3], a critical point for the dental sector.

For secondary use, the draft provides for **regional health data access bodies and a national body**, whose functions will be taken on by the Ministry of Health, coordinated with each other and integrated into the European HealthData@EU infrastructure [4]. This multi-level design is consistent with the division of powers, but it means a dental data space must deal with several bodies.

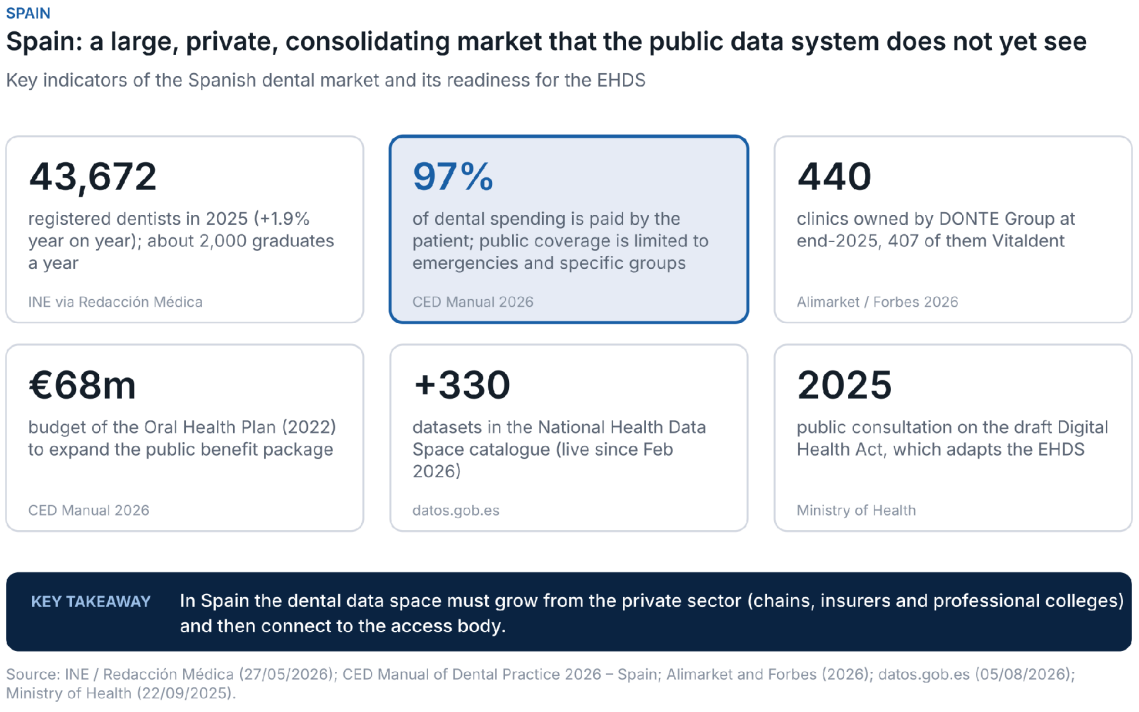
At the date of this study, the sources consulted do not record approval of the draft by the Council of Ministers. The reference framework therefore remains the GDPR and Organic Law 3/2018 on data protection, Law 41/2002 on patient autonomy and regional health record legislation.

3.5. Approach to oral health data

Spain has no national oral health register comparable to Denmark's. Population information comes from periodic surveys and regional child dental care programmes, whose data stays with the regional health services. Private data —by far the most abundant— is fragmented across thousands of clinics and several dozen chains and insurers, each with its own software. In practice, the approach is **sectoral and private**, with no public piece to bring it together.

Table 2 · Spain fact sheet

Indicator	Value	Source
Registered dentists (2025)	43,672 (+1.9%); 59.9% women	[20]
Active dentists (2024)	41,558 · 87.3 per 100,000 inhabitants	[2]
Dental spending paid by the patient	97%	[2]
Oral Health Plan (2022)	€68M; benefits extended to ages 0-14, pregnant women and other groups	[2]
Leading chain	DONTE Group: 440 owned clinics (407 Vitaldent) at end of 2025	[22]
National Health Data Space	Portal operational since Feb. 2026; 330+ datasets	[3]
EHDS adaptation law	Draft Digital Health Bill; public consultation from 22/09/2025	[4]
Planned access bodies	One national (Ministry of Health) and regional ones	[4]



Infographic 7 · *Spain: key indicators of the dental market and its readiness for the EHDS*

3.6. Market needs and regulatory needs

Market needs

- **Differentiation through quality.** In a market at risk of oversupply and with strong price competition, chains and insurers need comparable quality indicators across clinics to defend their value proposition and their relationship with patients.
- **Trust after the crises of large chains.** The sector still carries the reputational impact of the collapse of Dentix, some of whose clinics were taken over by Vitaldent [22]; transparency on clinical outcomes is a commercial asset.
- **Network management for insurers.** Insurers with their own or contracted dental networks need consistent data to manage costs and outcomes, without centralising identifiable records.
- **Data for innovation.** Manufacturers of implants, materials and artificial intelligence solutions need real-world evidence that they cannot currently obtain in an orderly way.

Regulatory needs

- **Inclusion of the private sector** in the interoperable health record envisaged by the Digital Health Act, with a format and timeline that dental software can meet.
- **Decision on Article 50**: whether or not Spain extends data holder obligations to microenterprises; in either case an aggregated participation channel is needed.
- **Coordination between regional and national bodies** so that a dental dataset from a chain operating in several regions can be requested only once.

3.7. Value of a dental data space and opportunity for partners

Spain is the market where a federated oral health data space adds the most distinctive value: the data exists, it is abundant and it has owners with incentives to share it under control, but no infrastructure brings it together. The recommended sequence is to start with the **internal use case of a chain or insurer** —comparable quality indicators across its clinics—, which funds the deployment of the connector, and then extend the federation to independent clinics through professional colleges or insurer networks. Once the national and regional bodies are operational, the dental space can be presented as a catalogued source.

For Microsoft Azure partners, the proposition fits Azure's penetration of the Spanish private healthcare sector and the requirement for data residency in the EU. Software in a Box brings experience from the SEDIA-funded data space demonstrators —in particular Global Data Care, in health— and its own Eclipse-based connectors.

4. Belgium

4.1. Health system and funding of dental care

Belgium has compulsory health insurance covering 99% of residents, funded mainly by payroll contributions and managed through sickness funds that apply common rules, fees and reimbursements [23]. The National Institute for Health and Disability Insurance (INAMI/RIZIV) is the central piece. Unlike Spain, **a significant share of dentistry goes through public reimbursement**: INAMI reimbursed just over €1 billion in dental care in 2021 and **€1.6 billion in 2025**, an increase of more than 50% against 16% growth in the number of treatments [5].

The relationship between dentists and sickness funds is organised through national dento-mutualist agreements that set maximum fees. The 2026-2027 agreement, signed in December 2025 and published in February 2026,

has been joined fully or partially by **60.48% of dentists**, compared with 56.97% for the previous agreement [24]; in 2024, 50.4% of the 7,869 registered practising dentists had rejected the agreement then in force [5].

4.2. Structure of the dental clinic market

The CED puts the number of **active dentists at 8,479** in 2023, equivalent to 71.7 per 100,000 inhabitants [23]. General dental practice requires working within the INAMI system after one year of vocational training, with places limited to 170 by federal decree [23]. The country has 514 active orthodontists and 205 periodontists and only 452 dental hygienists, a profession still developing [23].

Consolidation is more recent than in Spain or the Netherlands, but it has reached the regulatory agenda: the Dental Council (Conseil de l'Art dentaire / Raad van de Tandheelkunde) issued a specific opinion in November 2025 on commercial dental chains and their effect on the profession [25]. European groups such as Colosseum Dental operate in the Benelux.

4.3. Digital health infrastructure

Belgium has a federal eHealth platform underpinning online health services and a citizen portal (MaSanté / Mijn Gezondheid). The 2026-2029 interfederal eHealth action plan, successor to the 2025-2027 plan, focuses on operationalisation and synergies between projects [26]. For dentists, INAMI has granted since 2016 a **telematics bonus** linked to the use of telematics and electronic record management [26], which has encouraged the digitalisation of practices.

4.4. Regulatory framework and adaptation to the EHDS

The **Health Data Agency (HDA)** was created by the Act of 14 March 2023, in force since 13 April 2023, and was officially launched on 17 January 2024 [6]. Its mission is to facilitate the secondary use of health data in a uniform, transparent and secure way. The HDA has built a **national metadata catalogue with more than 480 datasets** and a data access application management system (DAAMS), and accepts requests for catalogued data and open requests for data not yet listed [6].

The HDA is expected to act as the **coordinating HDAB** and national contact point, as an interfederal institution and in collaboration with the Information Security Committee as the permit-granting entity; the designation still requires political approval at interministerial level [6]. The agency itself acknowledges that it still has to open dialogue with independent providers and smaller organisations, and that the cross-border connection to HealthData@EU will not be activated before March 2029 [6].

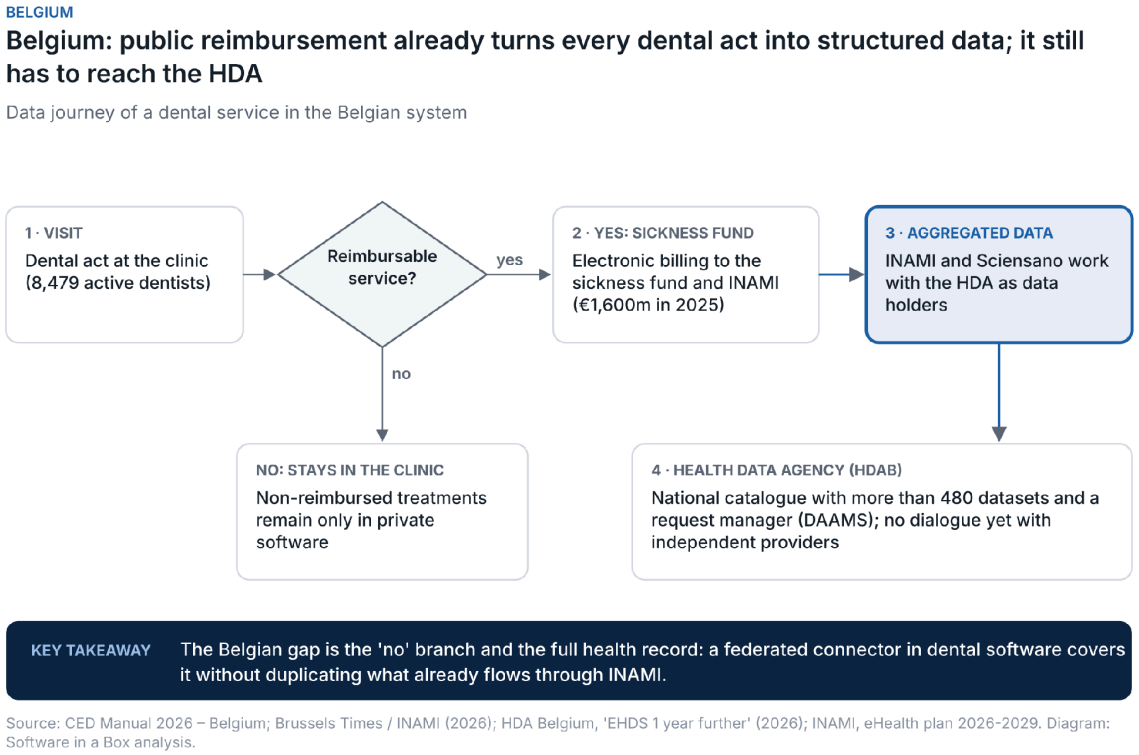
4.5. Approach to oral health data

The Belgian approach is that of a **payer**: the best-structured dental data is generated by billing reimbursable services to the sickness funds and INAMI. That data is valuable for reimbursement policy, but it does not capture the full health record or non-reimbursed treatments. The last national oral health data collection took place in 2012-2014 and has not been repeated since [27], leaving an epidemiological gap.

Table 3 · Belgium fact sheet

Indicator	Value	Source
Active dentists (2023)	8,479 · 71.7 per 100,000 inhabitants	[23]
Compulsory insurance coverage	99% of residents	[23]
INAMI reimbursement for dental care	≈€1,000M (2021) → €1,600M (2025)	[5]

Indicator	Value	Source
Adherence to the 2026-2027 dento-mutualist agreement	60.48% of dentists	[24]
Health Data Agency	Law of 14/03/2023; launched on 17/01/2024	[6]
HDA national catalogue	480+ datasets and DAAMS system	[6]
Last national oral health data collection	2012-2014	[27]
eHealth plan	Interfederal action plan 2026-2029	[26]



Infographic 8 · *Belgium: data journey of a dental service*

4.6. Market needs and regulatory needs

Market needs

- **Evidence on spending.** Dental reimbursement growing faster than treatment volume requires data explaining the evolution of cost per patient and per type of treatment.
- **Up-to-date epidemiological surveillance**, given the absence of a national oral health survey since 2014.
- **Positioning of chains** in the face of scrutiny by the professional regulator: transparency on outcomes is a response to the debate opened by the 2025 opinion.

Regulatory needs

- **Formalise the governance** of the HDA as HDAB, which the agency itself identifies as a priority.
- **Bring independent providers** —including dental practices— into the data holder ecosystem, with low administrative burden mechanisms.
- **Develop intermediary entities**: the HDA is exploring hospital data networks as intermediaries and envisages trusted data holders to ease the burden on the HDAB, following the experience of Findata in Finland [6].

4.7. Value of a dental data space and opportunity for partners

In Belgium the federated space does not compete with INAMI: it completes the **'no' branch**, that is, the non-reimbursed treatments and the full health record that today remain in each practice's software. The value proposition matches the HDA's stated need to reach independent providers and to have intermediaries and trusted holders. The recommended channel is a joint project with the HDA and bilingual professional associations, delivered by Azure integrators present in Brussels and Flanders. Software in a Box has an office in Brussels, in the European quarter, which makes this dialogue easier.

5. The Netherlands

5.1. Health system and funding of dental care

The Netherlands has a decentralised system with compulsory basic insurance. Dental care for **under-18s is included in basic insurance** —check-ups, cleaning, fluoride, periodontal treatment, root canals, fillings and some surgical treatments—, while adults mostly rely on voluntary supplementary insurance [28]. The public system covers **33% of the cost of dental care** (OECD, 2023), compared with 83% of total health spending covered by the State and social insurance [28]. Maximum fees are set by the health authority NZa.

5.2. Structure of the dental clinic market

In January 2025 the country had **9,555 active dentists**, 53 per 100,000 inhabitants, the lowest density in the group together with Germany [28]. The shortage is structural: according to the professional association KNMT, too few dentists have been trained for years, and around **a quarter of practising dentists trained abroad**; there are 2,287 dentists with a foreign degree, and in Zeeland they are more than half [7]. About **30% of practices do not accept new patients** [8].

The second force is **consolidation**. A 2025 study by the association Bevlogen Tandartsen identified **more than 500 practices owned by private equity**, around 13% of the total, with an estimated 30% share of revenue, and more than 21 active chains [8]. ABN AMRO estimated that 13% of some 4,400 practices belonged to a chain [8]. The competition authority ACM reviews deals such as the integration of Top Mondzorg and other groups, and Parliament has debated motions on private equity in healthcare [8].

5.3. Digital health infrastructure

The reference legislation is the **Wegiz** (Electronic Data Exchange in Healthcare Act), which requires certain prioritised datasets to be exchanged electronically according to quality and information standards. Its priority exchanges —basic data set (BgZ), urgent care, medication transfer, image availability and nursing transfer— resemble the EHDS categories, but are not identical [9]. Nictiz is the national competence centre for standards.

Dutch legal analysis itself acknowledges a bottleneck: the country **does not yet have a fully functioning national health information system**, and the national coverage network (LDN) is central to connecting to the EHDS [9]. Oral health is not among the prioritised Wegiz exchanges.

5.4. Regulatory framework and adaptation to the EHDS

Adaptation is organised through the **Wet GIS** (Health Information System Act), the first phase of a future framework law on data processing in healthcare. Its public consultation ran from 28 May to 8 July 2026 [9]. The

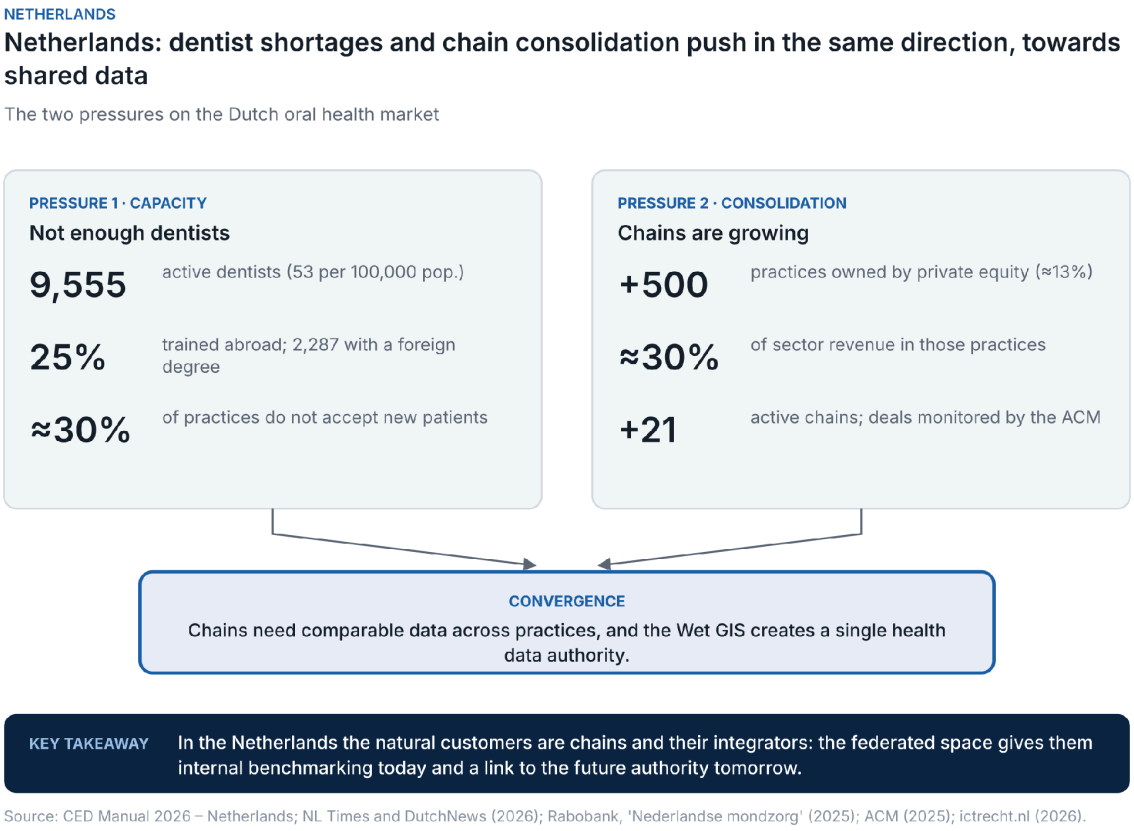
law designates the authorities required by 26 March 2027 and sets up the supervisory framework; primary use, secondary use and EHR systems will be regulated in a second phase [9].

The main institutional innovation is the creation of a **Gezondheidsdata Autoriteit** (Health Data Authority), an independent body that will take on the roles of digital health authority for primary use and HDAB for secondary use, as well as national contact point [9]. Some lawyers question its independence, since it will sit within the Ministry of Health (VWS), which is also a data holder and data user [9]. The law also replaces third-party certification of EHR manufacturers with self-assessment [9].

In parallel, **Health-RI** —a foundation dedicated to the reuse of health data through a national infrastructure— has been registered by the ACM as a data intermediation service under the Data Governance Act [29], a useful precedent for private federated models.

5.5. Approach to oral health data

The Dutch approach is that of a **regulated market**: fees set by the NZa, competition monitored by the ACM and a strong role for private providers organised in chains. Dental data has no public register of its own and no specific Wegiz exchange; it resides in practice software and in the systems of chains and insurers.



Infographic 9 · *The Netherlands: the two pressures on the oral health market*

Table 4 · The Netherlands fact sheet

Indicator	Value	Source
Active dentists (2025)	9,555 · 53.0 per 100,000 inhabitants	[28]
Public coverage of dental costs (2023)	33%; under-18s in basic insurance	[28]
Foreign-trained dentists	≈25%; 2,287 with a foreign degree	[7]

Indicator	Value	Source
Practices not accepting new patients	≈30%	[8]
Private equity practices	500+ (≈13%), ≈30% of turnover; 21+ chains	[8]
EHDS adaptation law	Wet GIS; consultation 28/05-08/07/2026	[9]
Authority and HDAB	Gezondheidsdata Autoriteit (primary and secondary use)	[9]
Registered data intermediary	Health-RI (Data Governance Act)	[29]

5.6. Market needs and regulatory needs

Market needs

- **Productivity with fewer professionals.** The shortage requires work to be reorganised between dentists, hygienists (about 3,900 in 2022) and assistants [28]; activity and outcome data are essential to do so with quality.
- **Post-acquisition integration.** Chains buying practices with heterogeneous software need to compare indicators without migrating all records at once.
- **Transparency towards the regulator and Parliament**, which are closely following the effect of private equity on costs and access.

Regulatory needs

- **Add oral health** to the data exchange agenda (Wegiz and EHDS), currently focused on hospital, primary and pharmaceutical care.
- **Guarantee the operational independence** of the new health data authority.
- **Develop the second phase** of the law (secondary use and EHR systems) in time for 2029.

5.7. Value of a dental data space and opportunity for partners

The Netherlands is, together with Spain, the priority opportunity market. The natural customers are **private equity-backed chains** and their technology integrators: the federated space offers them benchmarking across practices today, without costly migrations, and a route to connect to the Gezondheidsdata Autoriteit tomorrow. The precedent of Health-RI as a registered data intermediary makes it easier to explain the model to regulators. For selling through partners, a Dutch version of the sales material and the presentation video is a requirement.

6. Denmark

6.1. Health system and funding of dental care

Denmark has a national health service funded by general taxation. In dentistry the system is dual. **Children and young people receive free dental care** from the municipal dental service; coverage, which spanned ages 0 to 20 from 2023, was extended to **ages 0-21 from 2025** [10]. Adults, by contrast, receive a public subsidy of only about **15%**; adult dental care accounts for 31% of all out-of-pocket health payments in the country, and 42% of the population has supplementary insurance that can cover up to half of the remaining cost [10].

6.2. Structure of the dental clinic market

In 2025 Denmark had **4,964 active dentists** —8,808 registered—, i.e. 83.2 per 100,000 inhabitants, and 2,471 dental hygienists in 2022 [10]. According to the OECD, it is the only EU country where the number of dentists per inhabitant fell over the period analysed in its 2024 report [30]. Child care is mostly provided in municipal clinics, often located in schools, with public subsidy for care in private practices where there are no public clinics [11].

Adult practice is private and regulated by the dental agreement (tandlægeoverenskomst) negotiated between the regions and the dental association, which includes **ownership restrictions** that limit the expansion of chains [31]. Even so, chains have grown: by 2022 one in three adults was already using a dental chain [31], and Colosseum Tandlægerne, part of the European Colosseum group, operates around 26 clinics in the country.

6.3. Digital health infrastructure

Denmark is a European reference in digital health, with the sundhed.dk portal and a long tradition of national registers. The health reform has driven a new organisation: since **1 January 2026 the preparatory entity Digital Sundhed Danmark has been operating**, bringing together sundhed.dk, MedCom, the Danish Health Data Authority (Sundhedsdatastyrelsen) and various regional IT tasks [32]. Denmark is connecting to the European eHealth Digital Service Infrastructure (eHDSI) as part of its EHDS preparation [32].

For secondary use, a **Health Data Gateway** is being developed: a single entry point to data from registers, biobanks and clinical sources, supported by a DKK 200 million grant from the Novo Nordisk Foundation; the sources consulted describe it as still in development [33].

6.4. Regulatory framework and adaptation to the EHDS

Dentistry is regulated by the Danish Health Authority (Sundhedsstyrelsen), which administers the legislation, and by the Danish Patient Safety Authority, which grants authorisations and recognises specialists [10]. Institutional adaptation to the EHDS is taking place through the reform of digital governance —Digital Sundhed Danmark— and Sundhedsdatastyrelsen's projects on MyHealth@EU and secondary use [32]. The sources consulted do not yet confirm the formal designation of the Danish HDAB, so this study treats it as a process under way. Denmark also takes part in Nordic cooperation on the secondary use of health data (VALO project) [32].

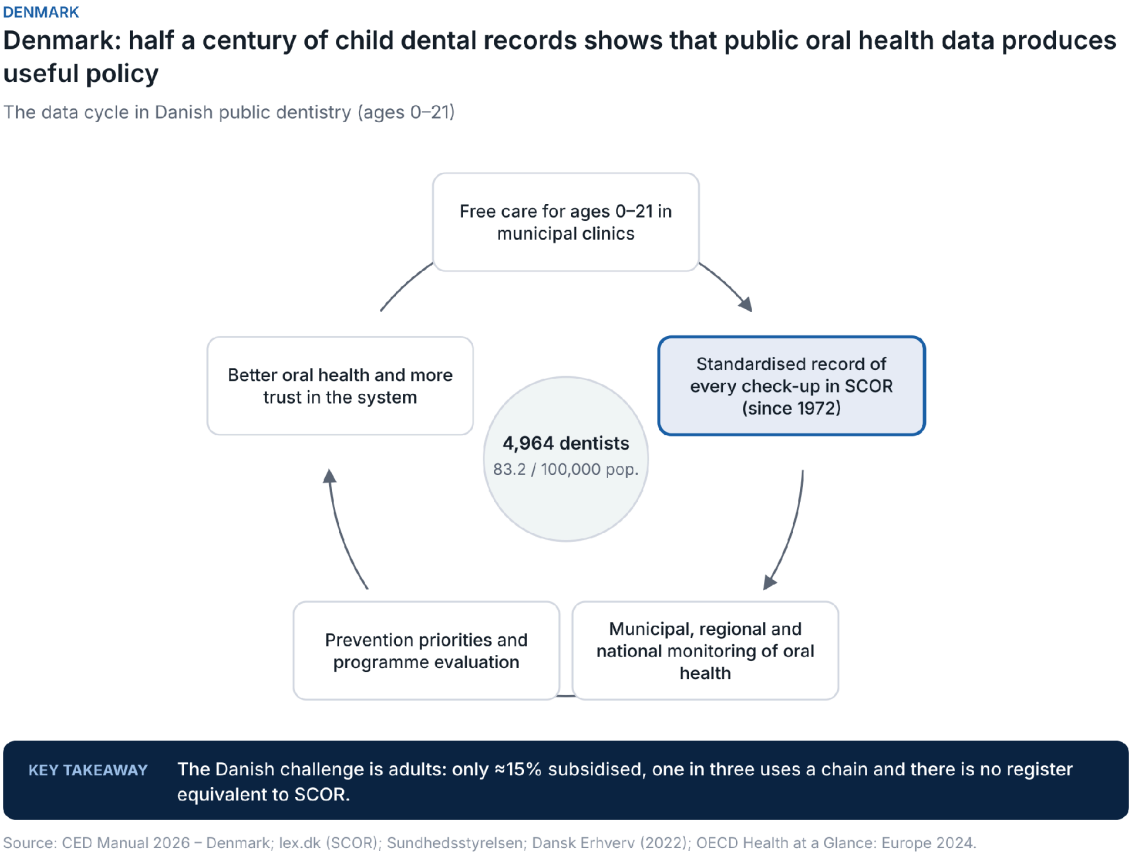
6.5. Approach to oral health data

Denmark is the only country in the group with a **long-standing national oral health register**: SCOR (the Danish Health Authority's Central Odontological Register), created with free child dentistry in 1972, which systematically records the check-ups of children and young people in a standardised way [11]. Municipal services are required to report their data to SCOR, and the register has made it possible to monitor oral health at municipal, regional and national level for more than five decades [11]. It is the best available demonstration that well-structured public oral health data produces useful policy.

Table 5 · Denmark fact sheet

Indicator	Value	Source
Active dentists (2025)	4,964 · 83.2 per 100,000 inhabitants	[10]
Free care	Ages 0-21 since 2025	[10]
Adult subsidy	≈15%; 42% with supplementary insurance	[10]
Dental hygienists (2022)	2,471	[10]
National dental register	SCOR, since 1972 (children and young people)	[11]

Indicator	Value	Source
Digital governance	Digital Sundhed Danmark (preparatory since 01/01/2026)	[32]
Data access	Health Data Gateway in development (Novo Nordisk Foundation, DKK 200M)	[33]
Adults treated in chains	One in three (2022)	[31]



Infographic 10 · *Denmark: the data cycle in public child dentistry*

6.6. Market needs and regulatory needs

Market needs

- **Visibility of adults.** There is no equivalent of SCOR for the adult population, treated in private practices and chains.
- **Capacity.** With falling dentist density, productivity and the allocation of tasks to hygienists require activity data.
- **Differentiation of chains** in a framework that limits their growth through ownership.

Regulatory needs

- **Complete the institutional transition** to Digital Sundhed Danmark and define the HDAB and the single data entry point.
- **Extend the SCOR logic to adults** without imposing disproportionate burdens on private practices, which requires voluntary or aggregated mechanisms.

6.7. Value of a dental data space and opportunity for partners

In Denmark the federated space does not have to prove the value of oral health data: SCOR has already done so. Its value lies in **bringing that logic to the adult population and private clinics**, connecting chains and practices through a federated model compatible with the new governance. The recommended channel is institutional —projects with Sundhedsdatastyrelsen or the regions— supported by Danish Azure integrators already validated by Software in a Box.

7. Germany

7.1. Health system and funding of dental care

Germany combines statutory health insurance (GKV), covering around 88% of the population, with substitutive private insurance (PKV) for 11% [34]. Dentistry is widely integrated into statutory insurance: **65% of dental spending is covered by statutory insurance** (2023), and dental spending equals 0.7% of GDP [34]. The National Association of Statutory Health Insurance Dentists (KZBV) defines the benefit catalogue together with the funds, and the German Dental Association (BZÄK) brings together the 17 regional chambers [34].

7.2. Structure of the dental clinic market

The CED puts the number of **active dentists at 43,663** in 2024 —out of 103,998 registered—, i.e. about 52 per 100,000 inhabitants [34]. The independent practice predominates, but consolidation is advancing through medical care centres (MVZ). According to the KZBV and BZÄK, **investor-owned MVZs (iMVZ) accounted for 30.4% of all dental MVZs** in the third quarter of 2023, with a rising trend; the profession is calling for limits on their expansion [35]. They are concentrated in large cities and in regions with patients willing to pay supplements.

7.3. Digital health infrastructure

In 2025-2026 Germany made the most visible leap in the group. The **'ePA für alle'** —an electronic health record for all statutorily insured people, with an opt-out model— started in pilot regions on 15 January 2025, was opened nationwide on 29 April 2025 and has been **mandatory for all dental practices since 1 October 2025** [12]. By April 2026 the funds had created an ePA for almost 73 million of the just over 74 million insured, there were **more than 5.5 million electronic dental bonus booklets (eZahnbonusheft)** accessible in the ePA, and up to 93,000 medical and dental practices were accessing their patients' ePA every week [12].

Dental telematics also has its own components, such as the electronic application and approval procedure (EBZ), in production nationwide since 1 January 2023 [36]. The KZBV stresses, however, that in dental practices the ePA is used mainly for medical history taking and that there are fewer mandatory document upload cases than in medicine [12].

7.4. Regulatory framework and adaptation to the EHDS

Germany has brought forward much of the secondary use architecture with the **Health Data Use Act (GDNG)**. The **Forschungsdatenzentrum Gesundheit (FDZ Gesundheit)**, the research data centre of the Federal Institute for Drugs and Medical Devices (BfArM), began accepting applications on 9 October 2025 and is designed as the piece that will connect Germany to the EHDS [13]. BfArM's **Data Access and Coordination Office (DACO/DAKO)** will act as coordinating HDAB, with domain-specific access bodies [13]. Connection of the ePA to the FDZ for secondary use of its data is planned for the end of 2026 [37].

On 15 July 2026 the Federal Government approved the draft **Act on Promoting Data and Digital Innovation in Healthcare (GeDIG)**, which sets the framework for data reuse across the entire health system and implements the EHDS building on existing infrastructure [37].

7.5. Approach to oral health data

The German approach is **institutional and integrated**: dental data flows through the public telematics infrastructure, KZBV billing generates detailed statistics —published annually in its Jahrbuch— and the ePA includes a dental component of its own, the eZahnbonusheft. It is the country where oral health data is closest to reaching the HDAB through the public route.



Infographic 11 · *Germany: value chain of dental data and contribution of a federated space*

Table 6 · Germany fact sheet

Indicator	Value	Source
Active dentists (2024)	43,663 · 52 per 100,000 inhabitants	[34]
Share of dental spending covered by statutory insurance (2023)	65%; GKV ≈88% of the population	[34]
ePA in dental practices	Mandatory since 01/10/2025	[12]
ePAs created / eZahnbonusheft	≈73 million / 5.5+ million (Apr. 2026)	[12]
EBZ procedure	In production since 01/01/2023	[36]
FDZ Gesundheit (BfArM)	Applications since 09/10/2025; DACO as coordinating HDAB	[13]

Indicator	Value	Source
EHDS adaptation law	GDNG in force; GeDIG approved by the Government on 15/07/2026	[37]
iMVZ as a share of all dental MVZs	30.4% (Q3 2023)	[35]

7.6. Market needs and regulatory needs

Market needs

- **Reducing the bureaucratic burden.** The KZBV complains that administrative tasks arising from regulation weigh on the daily running of practices [35]; any solution must save time, not add to it.
- **Management of iMVZ networks** under political scrutiny, with a need to demonstrate quality and contribution to care.
- **Data for software and industry**: dental practice software vendors must meet interoperability and conformity requirements.

Regulatory needs

- **Parliamentary passage of the GeDIG** and alignment of the coordinating HDAB with the domain bodies.
- **Effective ePA-FDZ connection** and definition of which dental contents are reused.
- **Regulation of iMVZs**, demanded by the profession.

7.7. Value of a dental data space and opportunity for partners

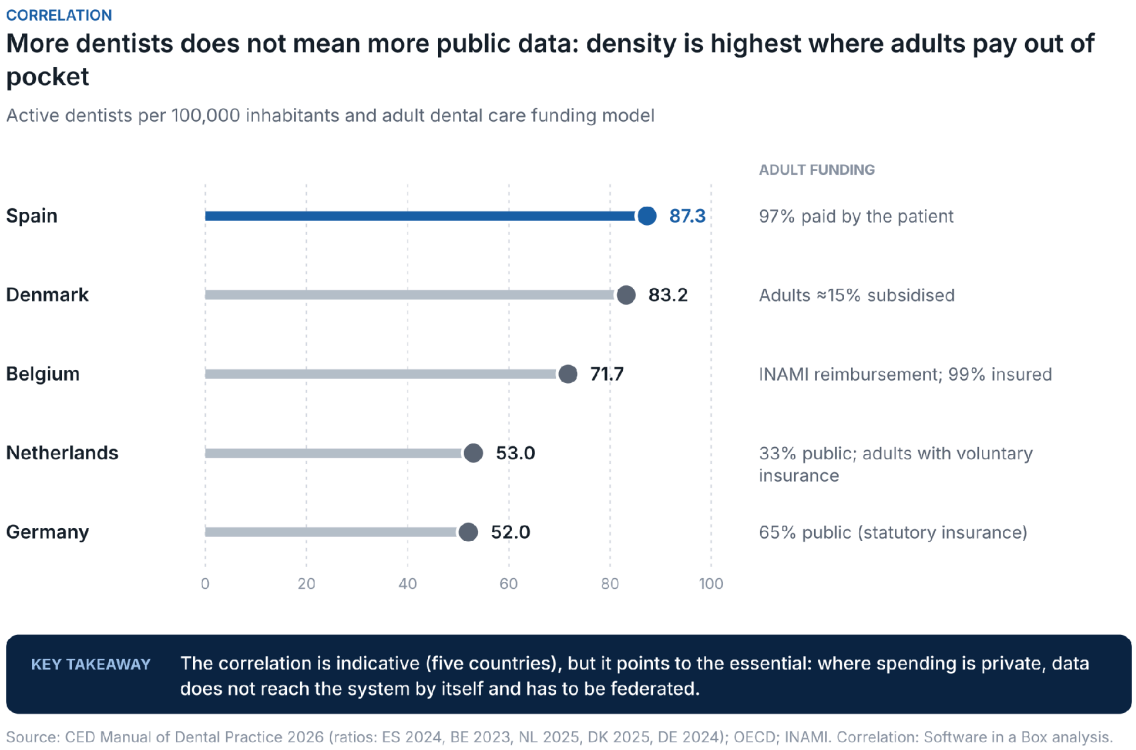
In Germany you do not compete with the ePA or the FDZ. The opportunity lies in the **missing pieces**: certifiable connectors for dental practice software, format and data quality validation before upload, federated analytics for chains and iMVZs that need to compare outcomes across centres, and catalogues with standardised metadata. The recommended channel is an agreement with a dental software vendor or an Azure integrator specialising in health telematics.

8. Comparative analysis

8.1. Correlation between professional density and public funding

Comparing the five countries reveals a striking relationship: **the countries with the most dentists per inhabitant are those where adults pay out of pocket**. Spain (87.3 per 100,000) and Denmark (83.2) lead on density and have, respectively, 97% private spending and an adult subsidy of around 15%; Germany (52) and the Netherlands (53) have the lowest density and, in the German case, the highest public coverage (65%). Belgium sits in between on both axes.

With only five observations, the relationship is indicative, not statistical. But its consequence for data architecture is clear: **where spending is private, dental data does not reach the public system by itself**, because it does not go through public billing or the system's health record. In those markets, a federated space is the condition for the data to exist at scale.



Infographic 12 · *Dentist density and adult funding model*

Table 7 · Comparison of dental clinic markets

Dimension	Spain	Belgium	The Netherlands	Denmark	Germany
Active dentists	41,558 (2024)	8,479 (2023)	9,555 (2025)	4,964 (2025)	43,663 (2024)
Per 100,000 inhabitants	87.3	71.7	53.0	83.2	52
Funding of dental care	97% out of pocket	INAMI reimbursement €1,600M (2025)	33% public	Free 0-21; adults ≈15%	65% public
Supply structure	Almost 100% private; chains and insurers	Contracted practice; debate on chains	Shortage; ≈13% PE-owned	Municipal for children; private for adults	Practice + MVZ; iMVZ 30% of MVZs
Key trend	Oversupply and consolidation	Rising spending	Consolidation and shortage	Falling density	Mandatory digitalisation
Approach to oral health data	Sectoral and private	Payer-driven	Regulated market	Public register	Institutional and integrated

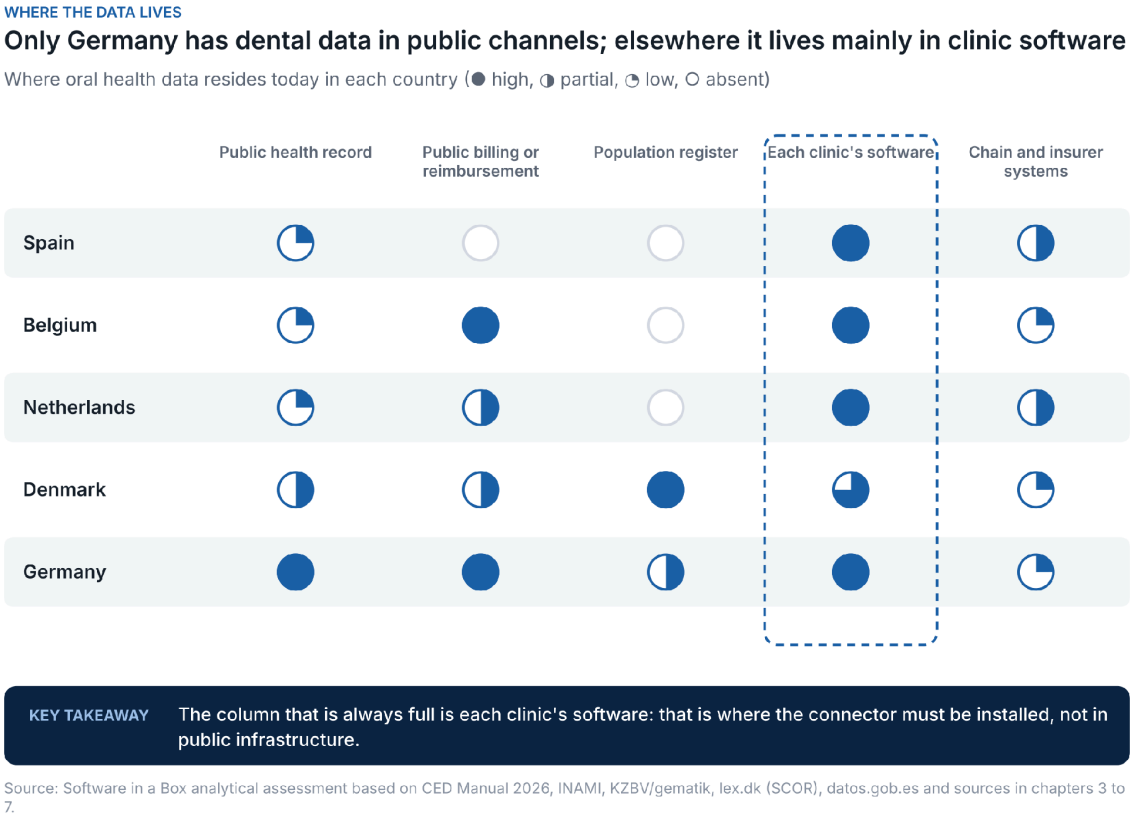
8.2. Differences in the approach to oral health data

Each country has developed a different approach to dental data, which shapes the form a data space must take:

- **Spain: sectoral and private.** Data is abundant but fragmented in private software; there is no public piece to bring it together.
- **Belgium: payer-driven.** The best-structured data is INAMI billing; the full health record and up-to-date epidemiological surveillance are missing.
- **The Netherlands: regulated market.** Chains and insurers concentrate the data; exchange regulation does not yet include oral health.

- **Denmark: public register.** SCOR has structured child data since 1972; adults are left out.
- **Germany: institutional and integrated.** Dental data flows through telematics and the ePA; the challenge is quality, reuse and the integration of iMVZs.

The practical consequence becomes clear when crossing countries and repositories: the only source of dental data present with intensity in all five markets is each clinic's practice software.



Infographic 13 · *Where oral health data resides today in each country (analytical assessment)*

8.3. Regulatory comparison

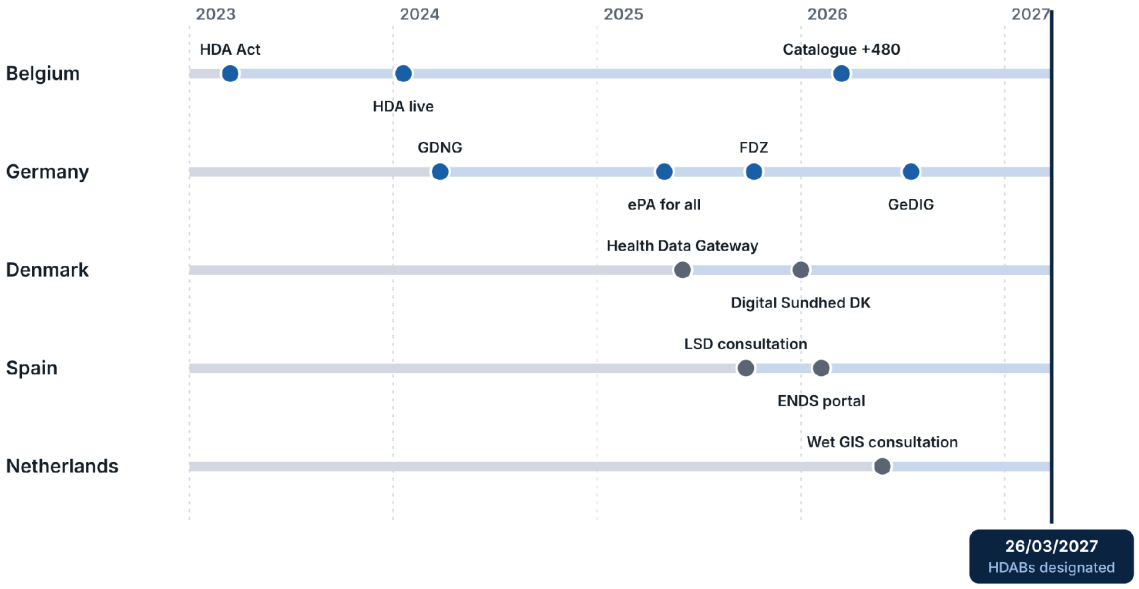
Table 8 · Comparison of health data regulation and governance

Element	Spain	Belgium	The Netherlands	Denmark	Germany
EHDS adaptation rule	Draft Digital Health Bill (consultation 2025)	HDA Law 2023; transposition pending	Wet GIS (consultation 2026)	Health reform and Digital Sundhed Danmark	GDNG; GeDIG (Government, 07/2026)
Secondary use body	National (Health Ministry) + regional	Health Data Agency (designation pending)	Gezondheidsdata Autoriteit	Sundhedsdatastyrelsen; Health Data Gateway	DACO (BfArM) + domain bodies
Operational infrastructure	ENDS: 330+ datasets	Catalogue of 480+ datasets; DAAMS	Health-RI; LDN network in development	National registers; eHDSI	FDZ Gesundheit operational
Health record / primary use	HCDSNS and ePrescription	eHealth platform; MaSanté	Wegiz; no complete national system	sundhed.dk	ePA for all (opt-out)
Dentist integration	Low (private sector)	Medium (INAMI billing)	Medium-low	High for children; low for adults	High (mandatory ePA)
Specific treatment of dental data	No	No	No	SCOR (children)	eZahnbonusheft in the ePA

NATIONAL TIMELINE

Belgium and Germany are years ahead; the Netherlands and Spain are running up against the March 2027 deadline

Milestones of institutional adaptation to the EHDS by country, to scale (2023-2027)



KEY TAKEAWAY Where the law arrives late (Spain, the Netherlands) the private sector has room to propose how dental data should be included.

Source: HDA Belgium and Hogan Lovells; Latham & Watkins; zm-online; Univ. Witten/Herdecke; Invest in Denmark; Altinget; Spanish Ministry of Health; datos.gob.es; ICTRecht.

Infographic 14 · *Milestones of institutional adaptation to the EHDS by country, 2023-2027*

The table and the timeline show that the five countries have chosen different institutional models for secondary use: a newly created specialised body (Belgium), a single authority for primary and secondary use (the Netherlands), an office within a medicines agency with domain bodies (Germany), a multi-level regional and national structure (Spain) and integration into a new national digital entity (Denmark). None of them, in the sources consulted, provides for specific treatment of oral health data.

8.4. Readiness map and opportunity matrix

To summarise the comparison, this study assesses six readiness dimensions on a scale of 1 to 4. This is an analytical assessment based on the sources in the previous chapters, not an official measurement.

Three readings stand out. First, **Germany is the only country with high readiness in almost every dimension**, which reduces the room for a new player but increases demand for compliance. Second, **Spain has solid public digital infrastructure but the lowest level of dentist integration and public funding**, creating a gap that can only be closed from the private sector. Third, **the structured dental register is the dimension with the lowest average and is at a low level in three of the five countries**: it is exactly the gap an oral health data space is meant to fill.

READINESS MAP

Germany arrives prepared; Spain has a digital public health record, but the dentist is left outside

Readiness for an EHDS-aligned dental data space by dimension (1 = low, 4 = high)

	EHR / national record	Dentist integrated	HDAB operational	National EHDS law	Dental register	Public funding
Spain	3 Advanced	1 Low	2 Partial	2 Partial	1 Low	1 Low
Belgium	3 Advanced	2 Partial	3 Advanced	2 Partial	1 Low	3 Advanced
Netherlands	2 Partial	2 Partial	2 Partial	3 Advanced	1 Low	2 Partial
Denmark	4 High	2 Partial	3 Advanced	2 Partial	4 High	3 Advanced
Germany	4 High	4 High	4 High	4 High	3 Advanced	4 High

KEY TAKEAWAY The structured dental register is the dimension with the lowest average (low in three of five countries): it is the gap an oral health data space is meant to fill.

Source: Software in a Box analytical assessment (scale 1–4), not an official measurement. Basis: sources cited in the country chapters.

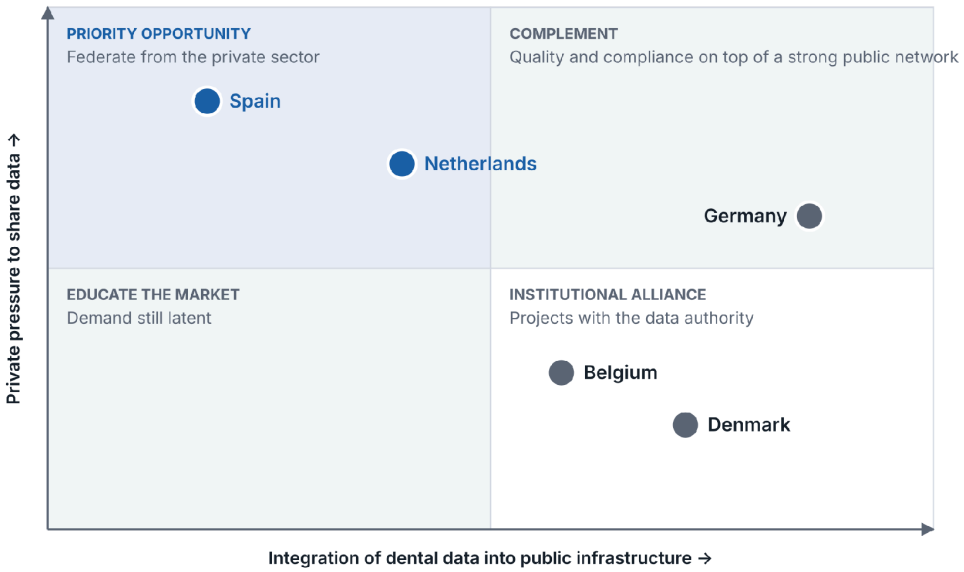
Infographic 15 · *Readiness map by country and dimension (analytical assessment)*

The matrix combines two axes: the degree of integration of dental data into public infrastructure and private pressure —consolidation, insurers, funds— to share and compare data. Spain and the Netherlands occupy the priority opportunity quadrant; Germany, the complement quadrant; Belgium and Denmark, the institutional alliance quadrant.

OPPORTUNITY MATRIX

Spain and the Netherlands are the markets where a private federated dental space adds the most value

Illustrative position of each country by public integration of dental data and private pressure to share data



DECISION — Recommended entry order: Spain and the Netherlands with chains and integrators; Germany as a quality and compliance complement to the ePA; Belgium and Denmark with the data authority.

Source: Illustrative, not measured, positions: Software in a Box analytical assessment based on the sources in chapters 3 to 7.

Infographic 16 · *Opportunity matrix: public integration versus private pressure to share data*

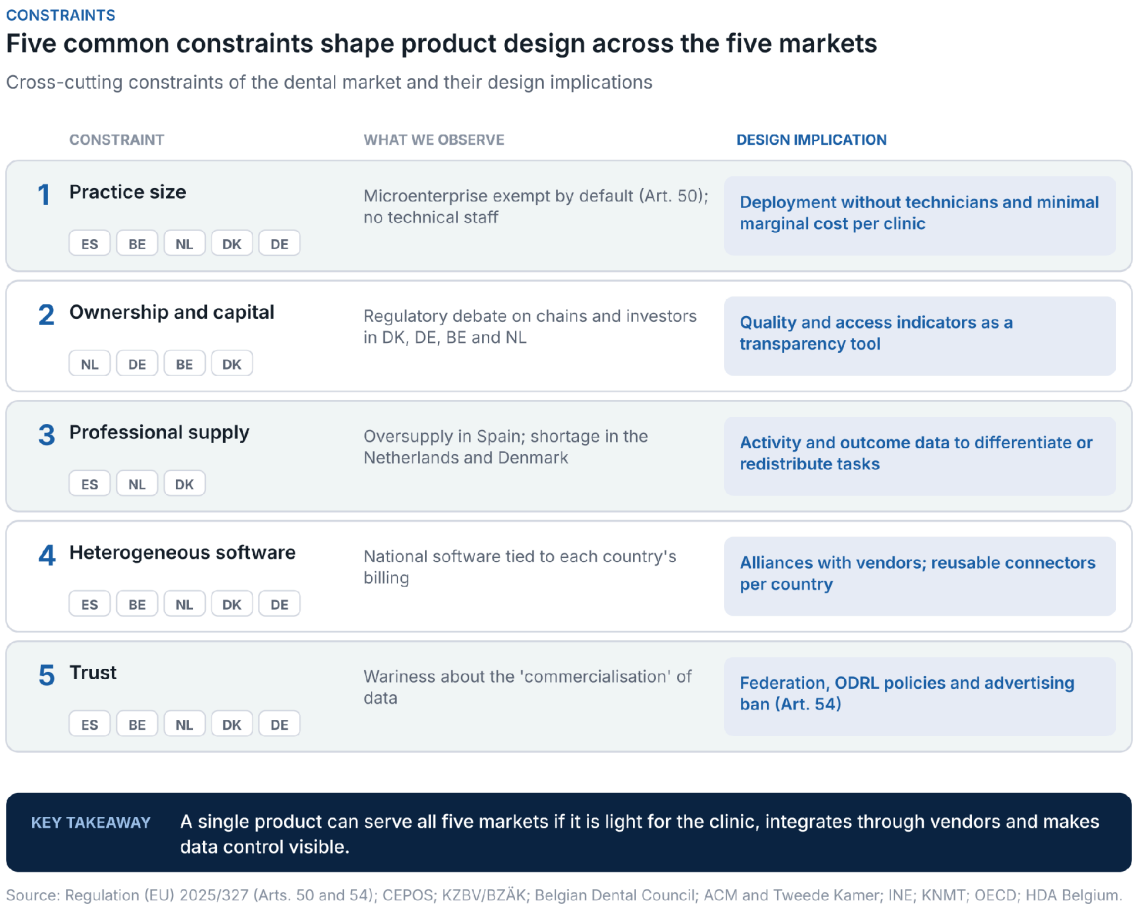
8.5. Cross-cutting market constraints

Beyond national differences, five constraints run across all five markets and must guide the design of any dental data space.

1. Practice size. The independent practice remains the dominant form of practice in all five countries. Its size exempts it by default from data holder obligations (Article 50) and limits its capacity to take on technology projects of its own. Any solution must be deployable without technical staff in the clinic and at a very low marginal cost per participating practice.

2. Regulation of ownership and capital. The entry of investors is the subject of regulatory debate in almost every market: Denmark keeps ownership restrictions in its dental agreement [31]; in Germany the profession is calling for limits on iMVZs [35]; in Belgium the Dental Council has issued an opinion on commercial chains [25]; in the Netherlands the competition authority reviews concentrations and Parliament debates the role of private equity [8]. A data space that brings transparency on quality and access can become an ally of chains in the face of that scrutiny.

3. Imbalance in professional supply. Spain faces a risk of oversupply, with about 2,000 graduates a year [20]; the Netherlands and Denmark face shortages and falling density [7][30]. In both scenarios activity and outcome data are needed: to differentiate in one case and to redistribute tasks between dentists and hygienists in the other.



Infographic 17 · *Cross-cutting constraints of the dental market and design implications*

4. Heterogeneous software. Each country has its own ecosystem of dental practice software, tied to national billing (sickness funds in Belgium, KZV in Germany, NZa fees in the Netherlands). That link to billing explains why national vendors dominate and why the connector must adapt to each one, preferably through alliances with the vendors themselves.

5. Professional and patient trust. The Belgian data agency identifies the perception of 'commercialisation' of data as a risk to public trust [6]. In dentistry, where the commercial relationship with the patient is direct, that perception is even more sensitive. Federated architecture, explicit usage policies and the Article 54 ban on advertising uses are the answer.

9. Market needs, regulatory needs and alignment with the EHDS

9.1. Market needs by country

Table 9 · Market needs by country

Country	Main need	Response of the dental data space
Spain	Comparable quality indicators across chain clinics and insurer networks; patient trust	Internal federation of the chain or insurer; extension to independent clinics via professional colleges
Belgium	Explain rising spending; epidemiological surveillance since 2014	Complete non-reimbursed data; oral health series for the HDA and INAMI

Country	Main need	Response of the dental data space
The Netherlands	Productivity under shortage; post-acquisition integration of practices	Benchmarking across practices without migrating records
Denmark	Visibility of adults treated privately and in chains	Extend the SCOR logic to adults with voluntary participation
Germany	Less bureaucracy; demonstrating iMVZ quality; software conformity	Data quality validation, federated analytics and connectors for vendors

9.2. Regulatory needs by country

Table 10 · Regulatory needs by country and link to the EHDS

Country	Regulatory need	Related EHDS mechanism
Spain	Include the private sector in the interoperable record; coordinate regional and national bodies; decide on microenterprises	Ch. III (EHR systems); Ch. IV (HDAB); Art. 50
Belgium	Formalise HDAB governance; bring in independent providers; intermediaries	Ch. IV; Art. 72 (trusted holders)
The Netherlands	Independence of the new authority; second phase of the law; include oral health in Wegiz	Digital health authority and HDAB; Ch. III
Denmark	Complete the institutional transition; define the HDAB and the single entry point	Ch. IV; HealthData@EU
Germany	Pass the GeDIG; ePA-FDZ connection; regulate iMVZs	Ch. IV; priority categories (Art. 14)

9.3. Alignment with Regulation (EU) 2025/327

The value of this analysis for selling through partners is that **every need identified already has an EHDS mechanism that supports it**. This makes it possible to present the dental data space not as yet another technology project, but as a route to early compliance.

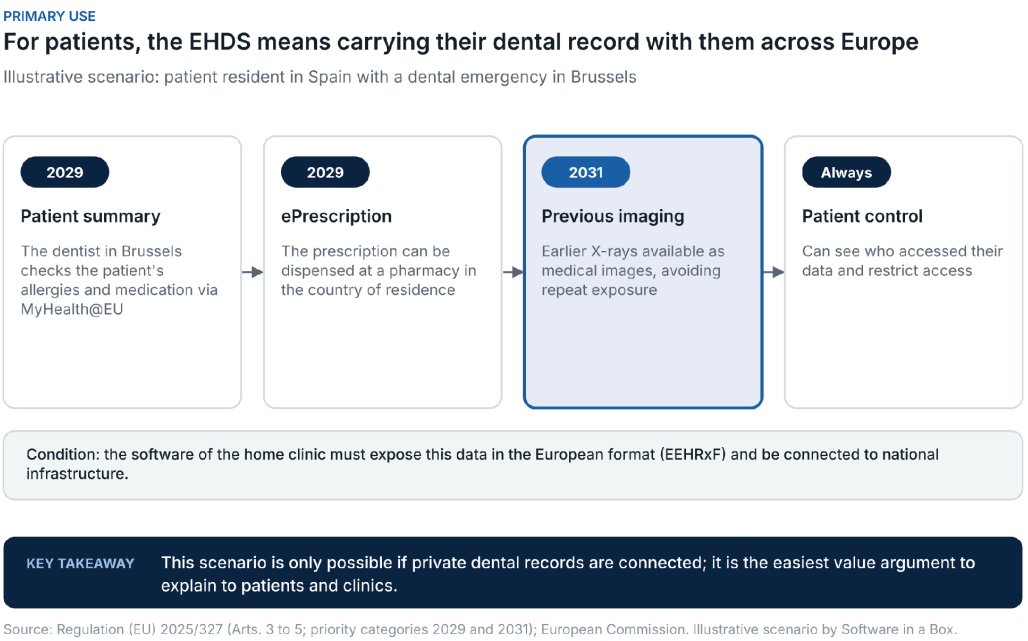
- **Interoperability of clinical software.** The Chapter III requirements on EHR systems —harmonised interoperability and logging components, the European EEHRxF format— give dental software vendors in all five countries a common horizon.
- **Patient rights.** Articles 3 to 5 —access, access services, insertion of information— turn the portability of the dental record between clinics, chains and countries into an enforceable right.
- **Secondary use with control.** Chapter IV allows data to reach researchers and innovators through HDAB permits and in secure environments, without holders handing over copies.
- **Inclusion of the small clinic.** Article 50 leaves the decision on microenterprises to each State; a federated space offers a voluntary, aggregated channel regardless of that decision.
- **Burden on chains.** Article 72 allows trusted data holders to be designated to handle requests concerning their own data.
- **Public trust.** Article 54 prohibits advertising or discriminatory uses, a key argument against the perception of 'commercialisation' of data that also concerns the agencies [6].
- **Dental imaging.** The inclusion of medical images in the priority categories in 2031 opens the way to the reuse of X-rays and CBCT.

The full mapping between needs and mechanisms is summarised in the following diagram.



Infographic 18 · *Mapping between dental market needs and EHDS mechanisms*

For patients, alignment with the EHDS has a very concrete meaning, which is also the easiest argument to explain to clinics:



Infographic 19 · *Illustrative primary use scenario: patient with a dental emergency in another Member State*

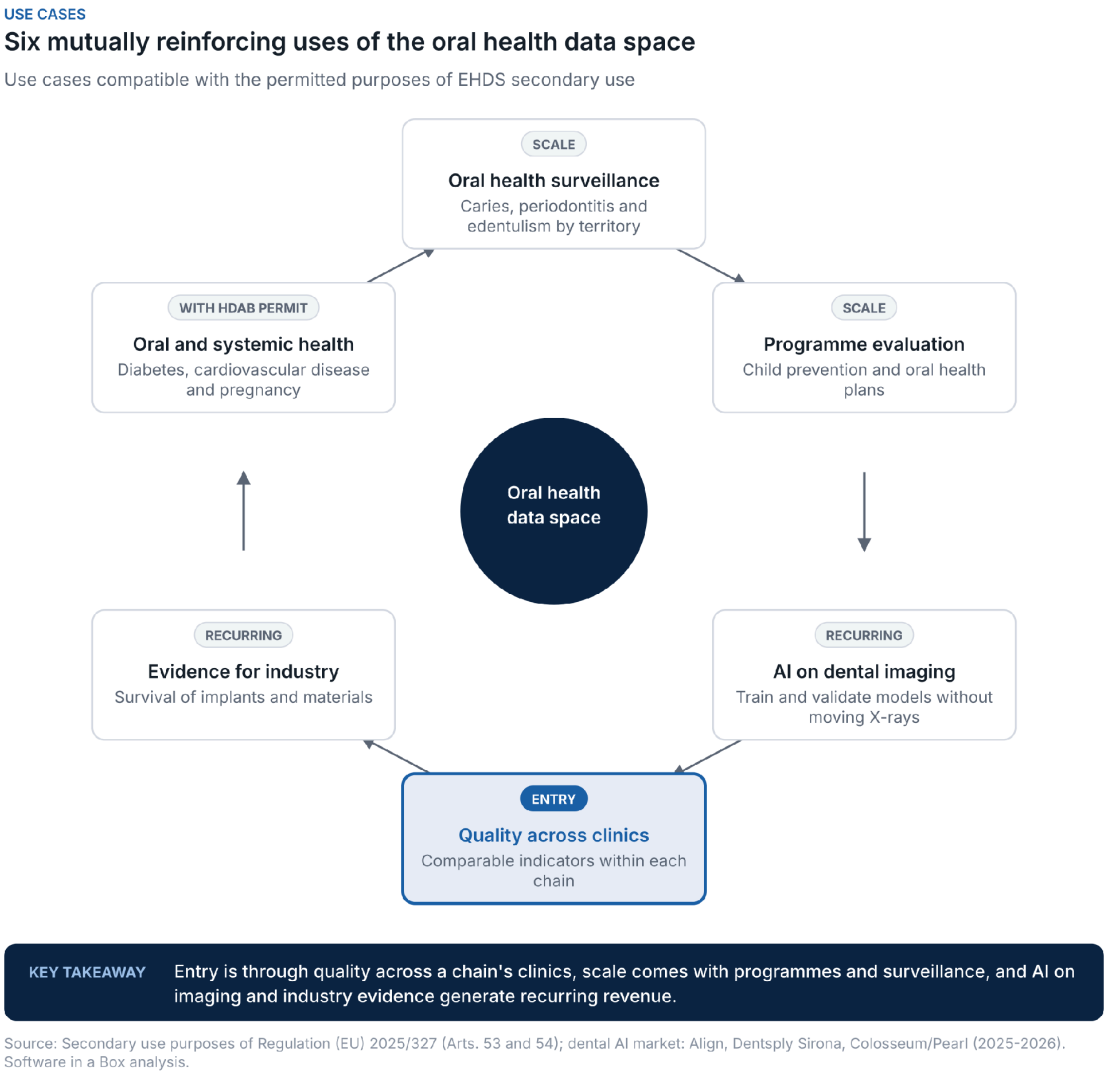
9.4. The importance of national participation in the EHDS

The comparison shows that the EHDS sets the framework, but **the real value for dental data is decided by national rules**: whether obligations are extended to microenterprises, whether the interoperable health record includes the private sector, whether the HDAB has the capacity to deal with independent providers, whether recognised intermediaries exist. In all five countries these decisions are being taken between 2025 and 2027. This is the moment when professional colleges, chains, insurers and technology providers can influence matters so that oral health is not left out.

> Positioning recommendation: in each country, Software in a Box and its partners should take part in the public consultations and working groups on EHDS adaptation with a concrete proposal for dental data —voluntary aggregated participation, trusted holders for chains and common metadata—, backed by a pilot already under way.

10. Use cases for the oral health data space

The use cases have been selected for their compatibility with the secondary use purposes permitted by the Regulation and their capacity to generate recurring revenue for partners.



Infographic 20 · *Six use cases for the oral health data space*

1. Quality across clinics. Comparable indicators —treatment success, re-treatments, waiting times, adherence to check-ups— calculated in a federated way across the clinics of a chain or an insurer network. This is the entry case: it has a paying customer, does not require an HDAB permit because the data is used within the group itself on its own legal basis, and prepares the infrastructure for everything else.

2. Artificial intelligence on dental imaging. The dental diagnostic AI market is growing rapidly in Europe: Align Technology launched in the EU an assisted detection solution for 2D X-rays; Dentsply Sirona announced in May 2026 a CE-marked tool to detect periapical radiolucencies in CBCT; and Colosseum Dental Group, with more than 630 clinics in 11 countries, is piloting AI diagnostic solutions [38]. Federated training and validation make it possible to improve these models without moving X-rays out of the clinics.

3. Programme evaluation. Child dental care programmes (Spain, the Netherlands, Denmark) and oral health plans need to measure outcomes with real data, beyond activity.

4. Oral health surveillance. Prevalence of caries, periodontal disease and edentulism by territory, particularly valuable in Belgium, with no national survey since 2014, and in Spain, with no national register.

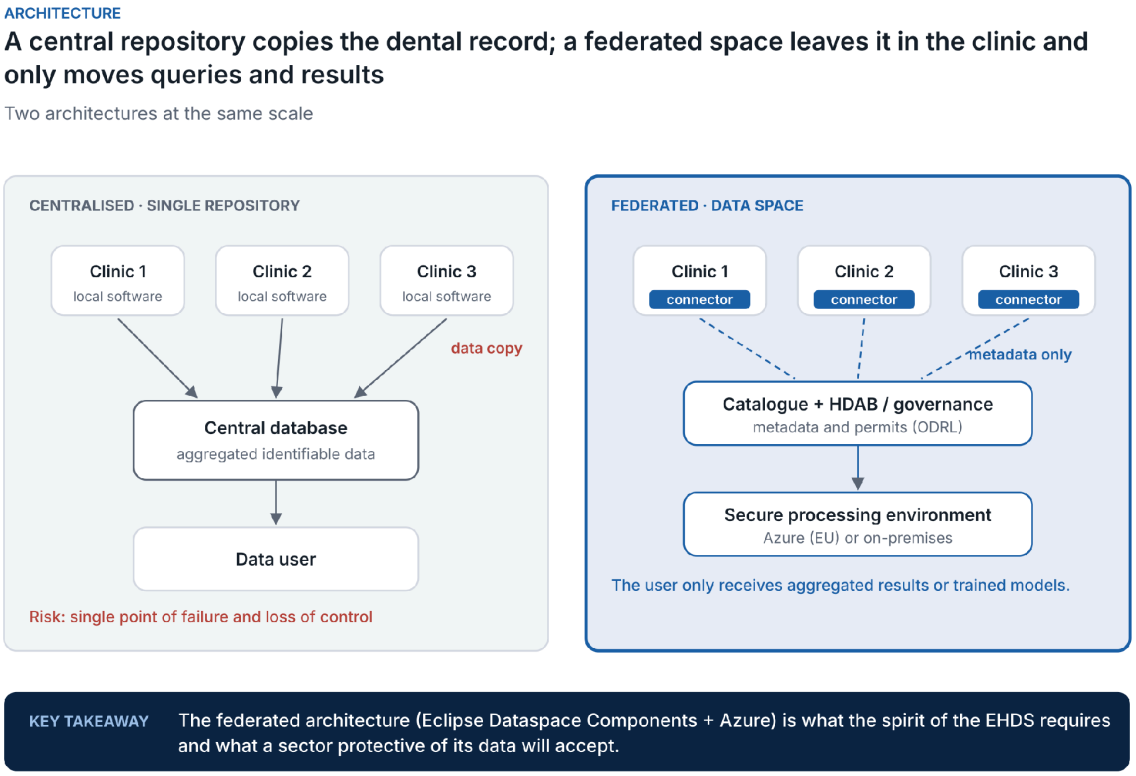
5. Evidence for industry. Survival of implants and performance of materials based on real-world data, within the permitted purposes of product development and innovation and without advertising uses.

6. Oral and systemic health. The relationship between periodontal disease and diabetes, cardiovascular disease or pregnancy, which requires crossing dental and medical data and is only feasible through HDAB permits.

11. Proposed architecture and the role of Software in a Box

11.1. Why a federated architecture

A central repository would copy identifiable dental records into a single database, with a single point of failure and a loss of control the sector would not accept. A federated data space leaves the data in each clinic and only moves metadata, permits and, where applicable, aggregated results or trained models. It is the model behind the EHDS itself —processing in secure environments, without copies— and the one already used, for example, by Spain's National Health Data Space [3].



Infographic 21 · *Centralised versus federated architecture*

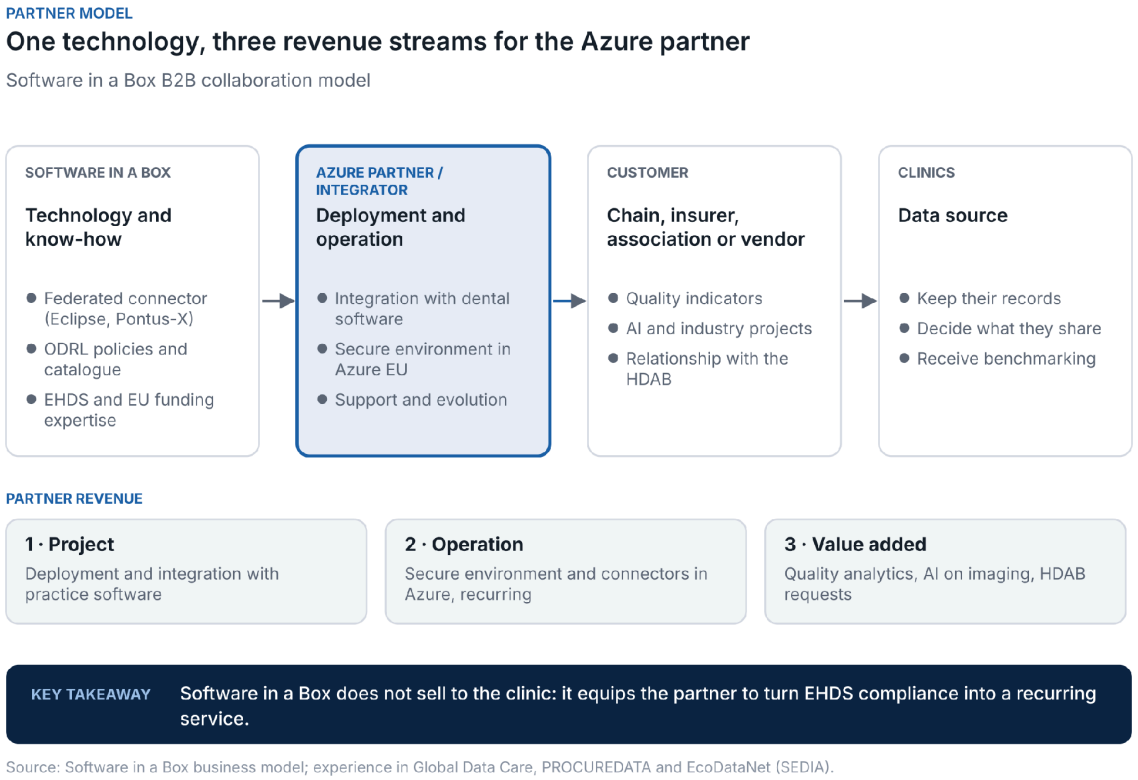
11.2. Components

- **Data space connector** based on Eclipse Dataspace Components, built into the practice software or deployed alongside it, which exposes catalogues and enforces usage policies.
- **Machine-readable usage policies (ODRL)** that translate HDAB permits and agreements between holders into executable rules.
- **Catalogue with HealthDCAT-AP metadata**, the profile HDABs use to describe datasets [3].
- **Secure processing environment** on Microsoft Azure in EU regions, or on the customer's premises when data sovereignty requires it.

- **Interoperability layer** towards the European EEHRxF format and HL7 FHIR standards for the priority categories.

11.3. Role of Software in a Box and its partners

Software in a Box does not intend to sell directly to clinics. Its model is **B2B**: it provides federated data space technology and regulatory know-how to Microsoft Azure partners, technology integrators and software developers, who build it into their solutions for chains, insurers, associations and clinics. Software in a Box's experience in the SEDIA-funded data space demonstrators —Global Data Care in health, PROCUREDATA in procurement and EcoDataNet in the circular economy— and in developing its own connectors on Eclipse and Pontus-X is its main competitive advantage.



Infographic 22 · *B2B collaboration model and partner revenue streams*

For the partner, the model offers three revenue streams: the deployment project and integration with the practice software; recurring operation of the environment on Azure; and value-added services on the use cases (quality analytics, AI on imaging, support for HDAB requests).

11.4. Compliance framework for the connector

The dental data space is not governed by the EHDS alone. Its design must incorporate from the outset a set of European rules that apply simultaneously:

- **General Data Protection Regulation (GDPR).** Health data is a special category (Article 9) and requires a specific legal basis, an impact assessment and enhanced measures [39]. The EHDS complements the GDPR; it does not replace it.
- **Data Governance Act (DGA).** When a third party brings data holders and users together, it may be a data intermediation service subject to notification and neutrality [40]. Health-RI's registration in the Netherlands

[29] shows how this applies to health; whether the operator of the dental space falls within that category should be analysed case by case.

- **Artificial Intelligence Act.** AI systems that are medical devices, or safety components of them, subject to third-party conformity assessment are considered high-risk [41]. Diagnostic AI on dental X-rays falls within that scope, with training data governance obligations that a federated space helps to meet.
- **Medical Devices Regulation (MDR).** Software with a diagnostic purpose is a medical device and requires CE marking [42]; the data space provides clinical evidence and post-market surveillance.
- **EHDS.** Requirements for EHR systems (Chapter III), data holder obligations and the permit procedure (Chapter IV), and prohibited uses (Article 54) [15].



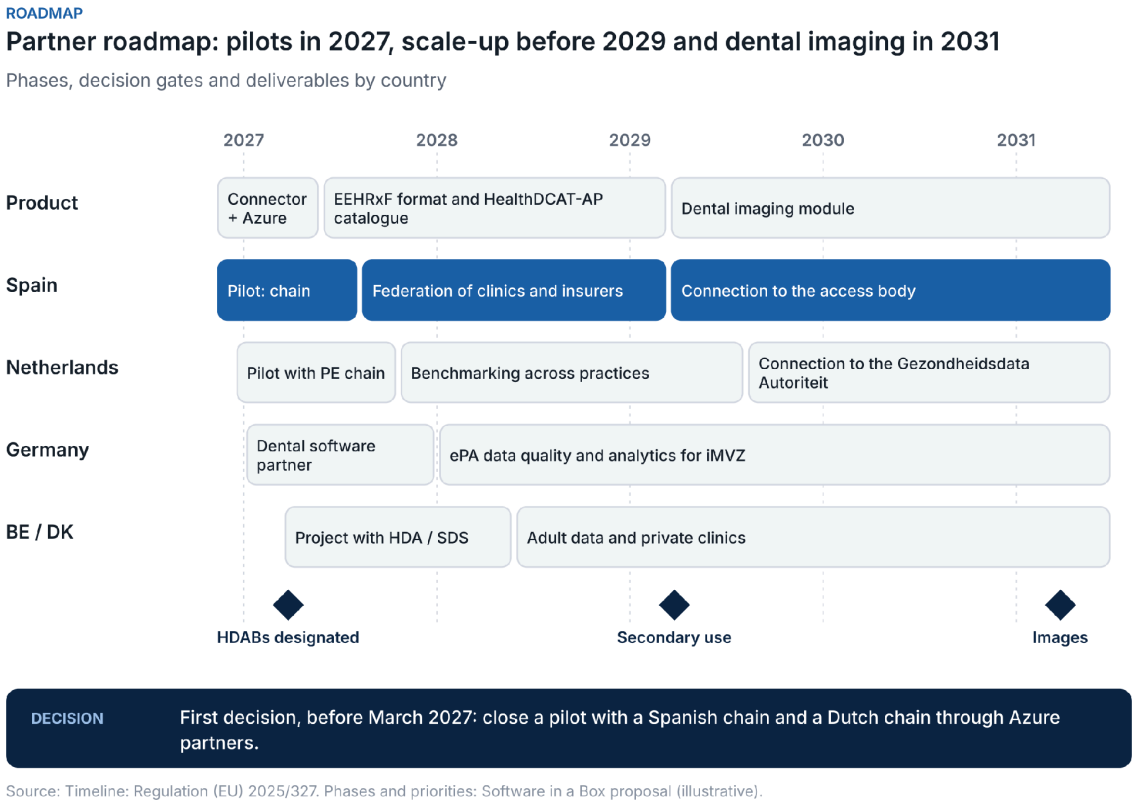
Infographic 23 · *European regulatory layers affecting the oral health data space*

Table 11 · Regulatory requirements and design response

Regulation	Relevant requirement	Response in the dental data space
GDPR [39]	Health data as a special category; legal basis, minimisation and impact assessment	Data at source; pseudonymisation in the clinic; impact assessment per use case
EHDS [15]	Interoperability of EHR systems; HDAB permits; secure environments; Art. 54	EEHRxF format; HealthDCAT-AP catalogue; secure environment on Azure (EU)
DGA [40]	Neutrality and notification of data intermediation services	Analysis of the operator's role; separation between intermediation and value-added services
AI Act [41]	High risk for AI embedded in medical devices; training data governance	Traceability of training datasets and federated validation
MDR [42]	CE marking and post-market surveillance of diagnostic software	Real-world clinical evidence from the federation

12. Roadmap 2026-2031

The roadmap is organised according to the EHDS timeline and the opportunity matrix. It is a Software in a Box proposal, subject to validation with partners and customers.



Infographic 24 · *Partner roadmap 2026-2031*

- **Phase 0 (November 2026 - March 2027): preparation.** Dental connector on Azure ready for pilots; agreements with partners in Spain and the Netherlands; sales material and videos by language; participation in public consultations.
- **Gate 1 (26 March 2027): designation of HDABs.** Decide in which countries to seek formal dialogue with the access body.
- **Phase 1 (2027-2028): pilots.** Quality across clinics with a Spanish chain and a Dutch chain; agreement with a software vendor in Germany; project with the Belgian HDA or with Sundhedsdatastyrelsen.
- **Gate 2 (26 March 2029): start of secondary use.** The dental space must be catalogued with HealthDCAT-AP metadata and with datasets ready for requests.
- **Phase 2 (2029-2031): scale-up.** Federation of independent clinics, trusted data holders, connection to the access bodies.
- **Gate 3 (26 March 2031): medical images.** Launch of the dental imaging module for exchange and reuse.

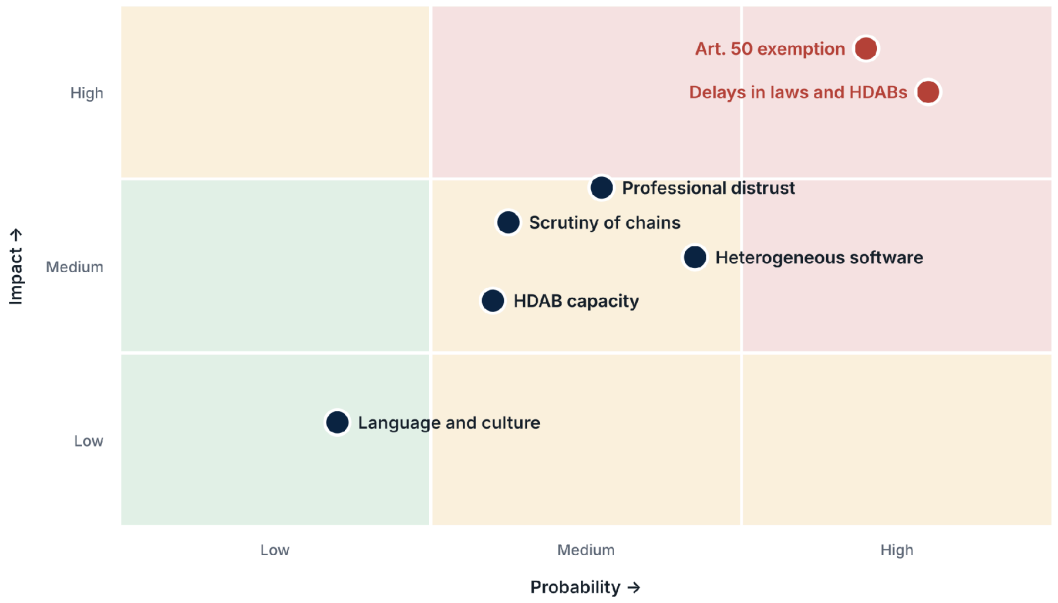
13. Risks and constraints

Deployment risks have been assessed by probability and impact. The two critical ones are regulatory in nature and share the same mitigation: start with use cases in which chains share data in their own interest, without depending on the access bodies' timeline.

RISKS

The two critical risks are regulatory and are mitigated by starting with private use cases

Probability and impact matrix of deployment risks (analytical assessment)



DECISION — Both critical risks (regulatory timeline and microenterprise exemption) are neutralised by starting with chains that share data in their own interest.

Source: Software in a Box analytical assessment; see the risk table in chapter 13.

Infographic 25 · *Probability and impact matrix of risks (analytical assessment)*

Table 12 · Deployment risks and constraints

Risk	Level	Mitigation
Delays in national laws and in the designation of HDABs	High	Start with chains' internal use cases, which do not depend on the HDAB; follow public consultations.
Microenterprise exemption (Art. 50) reducing the mandatory supply of data	High	Voluntary, aggregated participation model; trusted holders for chains.
Professional distrust of the 'commercialisation' of data	Medium	Federated architecture; ODRL policies; highlight the ban on advertising uses (Art. 54).
Heterogeneity of dental practice software	Medium	Enter through software vendors and partners; reusable connectors per country.
Independence and capacity of the access bodies	Medium	Offer well-catalogued datasets (HealthDCAT-AP) that reduce their workload.
Political scrutiny of chains and private equity (NL, DE, BE)	Medium	Position the space as a transparency and quality tool.

Risk	Level	Mitigation
Language and commercial culture in each country	Low	Material and videos per language; local partners.

14. Conclusions

Oral health data is one of the great untapped assets of European healthcare, and the EHDS offers for the first time a common framework to bring it in. But the comparison between Spain, Belgium, the Netherlands, Denmark and Germany shows that **the Regulation alone does not guarantee that dental data will arrive**: the structure of the sector —small practices, private spending, proprietary software— and the microenterprise exemption may leave it out.

The five countries start from very different approaches: sectoral and private in Spain, payer-driven in Belgium, regulated market in the Netherlands, public register in Denmark and institutional in Germany. These differences are not an obstacle but the basis for a differentiated strategy: **federate from the private sector** where data is fragmented (Spain and the Netherlands), **complement** where public infrastructure already channels it (Germany) and **ally with the data authority** where the challenge is to bring in adults and private clinics (Belgium and Denmark).

In every case, each country's participation in the EHDS and the decisions it takes in its national legislation between 2025 and 2027 will determine whether oral health becomes part of the European space. Software in a Box and its partners can help make that happen with a concrete proposal, a pilot under way and a federated architecture that respects professionals' control over their data.

About Software in a Box

Software in a Box is a technology company based in Madrid with an office in Brussels, specialising in federated data spaces, data sovereignty and proprietary artificial intelligence. Its model is exclusively B2B: it develops the technology and regulatory know-how that Microsoft Azure partners, technology integrators and software developers build into their solutions for end customers.

Its experience comes from the federated data space demonstrators funded by SEDIA and delivered between 2024 and 2026: **Global Data Care** (health), **PROCUREDATA** (procurement) and **EcoDataNet** (circular economy and waste). In them it has developed its own connectors on Eclipse Dataspace Components and Pontus-X, aligned with the Gaia-X and IDSA frameworks and with ODRL usage policies.

The oral health data space applies that experience to a specific sector, with an architecture on Microsoft Azure designed for partners to deploy with chains, insurers, professional associations and dental software vendors in Spain, Belgium, the Netherlands, Denmark and Germany.

> Partner programme: integrators and Microsoft Azure partners interested in adding the oral health data space to their offering can request the partner briefing at softwareinabox.eu or by writing to emilio.mulet@softwareinabox.eu.

Annex A. Stakeholder map by country

This annex identifies the stakeholders that appear in the analysis and their potential role in a dental data space. It is not intended to be exhaustive and should be completed with partner research in each country.

Table 13 · Relevant stakeholders by country

Country	Institutions and regulators	Private and professional sector
Spain	Ministry of Health (General Secretariat for Digital Health); regional health services; National Health Data Space; Ministry for Digital Transformation	General Council of Dentists and professional colleges; DONTE Group (Vitaldent, MAEX); insurers with dental networks (Sanitas, Adeslas, Asisa, DKV, Caser); Vivanta; Institutos Odontológicos
Belgium	INAMI/RIZIV; Health Data Agency; FPS Public Health; Sciensano; Information Security Committee; Dental Council	Sickness funds (mutualities); dentists' professional associations; Benelux chains (Colosseum Dental Benelux)
The Netherlands	Ministry of VWS; future Gezondheidsdata Autoriteit; NZa; IGJ; ACM; Nictiz; Zorginstituut	KNMT; private equity-backed chains (Top Mondzorg, Fresh, among others); health insurers; Health-RI
Denmark	Sundhedsstyrelsen; Styrelsen for Patientsikkerhed; Sundhedsdatastyrelsen; Digital Sundhed Danmark; municipalities and regions	Danish Dental Association (Tandlægeforeningen); chains (Colosseum Tandlægerne and others); Novo Nordisk Foundation (Health Data Gateway)
Germany	Federal Ministry of Health; BfArM (FDZ Gesundheit, DACO); gematik; statutory health insurance funds	KZBV and regional KZVs; BZÄK and regional chambers; iMVZs and groups (Colosseum Dental Deutschland); dental software vendors

Annex B. Partner FAQ

Is the dental clinic obliged to share data under the EHDS?

As a data holder for secondary use, a microenterprise is exempt unless its State decides otherwise (Article 50). As a healthcare provider, it must respect patients' access rights and, insofar as it processes priority categories, use conformant EHR systems [15][1].

Why isn't each country's public health record enough?

Because in Spain, and in much of adult care in the Netherlands and Denmark, dental data does not go through public infrastructure. Only Germany has made it mandatory for dental practices to be integrated into its electronic health record [12].

Does patient data leave the clinic?

Not in the federated model. The clinic keeps its data; the connector exposes metadata and enforces usage policies. Analyses run in secure environments and the user receives aggregated results or trained models.

What role does Microsoft Azure play?

Azure provides the secure processing environments, data residency in EU regions and health interoperability services. When data sovereignty requires it, the same model can be deployed on the customer's premises.

What is the recommended first project?

A pilot of quality indicators across the clinics of a chain or insurer network: it has a customer to fund it, it does not depend on the HDAB being operational and it leaves the connector installed for later use cases.

What happens in 2029?

The exchange of patient summaries and ePrescription begins, together with most secondary use rules [17]. A dental space catalogued with HealthDCAT-AP metadata will be able to receive requests through the access bodies.

And in 2031?

Medical images join the priority categories [18]. This is the time to offer the dental imaging module for exchange and for federated training of AI models.

Annex C. Glossary

Table 14 · Glossary

Term	Definition
EHDS	European Health Data Space, established by Regulation (EU) 2025/327.
Primary use	Processing of electronic health data to provide healthcare.
Secondary use	Reuse of health data for research, innovation, public policy, statistics or training.
HDAB	Health Data Access Body: the body that grants permits for secondary use.
Trusted data holder	Designated entity that handles access requests concerning only its own data (Art. 72).
EEHRxF	European Electronic Health Record Exchange Format.
HealthDCAT-AP	Metadata profile for describing health datasets in catalogues.
MyHealth@EU / HealthData@EU	European infrastructures for cross-border primary use and for secondary use.
Federated data space	Architecture in which data stays at source and is shared under usage policies through connectors.
ODRL	Standard language for expressing machine-readable data usage policies.
ePA	German electronic health record ('elektronische Patientenakte').
iMVZ	Investor-funded medical care centre in Germany.
SCOR	Danish central odontological register of child oral health.
ENDS	Spain's National Health Data Space.

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